Alternative moduli of twisted curves

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Abstract

We define new Artin stack compactifications of the moduli of smooth curves, which parameterize "relatively polarized curves" consisting of stacky semistable curves equipped with line bundles that are trivialized away from the rational bridges. Using techniques from birational geometry, we prove that stacks of relatively polarized curves admit the classical coarse space of stable curves as a good moduli space. One of our main motivations is the companion paper [HLHIS26], where we construct a proper moduli space of SL_N-bundles on relatively polarized curves. This suggests the hypothesis that relatively polarized curves are suitable sources for defining Gromov–Witten theory for Artin stack targets.

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1 Introduction

The first, and most fundamental, moduli problem studied in algebraic geometry was the moduli of smooth curves of genus g. In order to study the topology of that moduli space, one often compactifies it by adding nodal curves as in the Deligne-Mumford compactification $\overline{M}_g$ [DM69]. Likewise, in enumerative geometry, one compactifies the moduli of smooth curves embedded in a smooth and proper target variety in order to reduce enumerative questions to questions about intersection theory on moduli

spaces. In Gromov–Witten theory this compactification, Kontsevich's moduli of stable maps [Kon95], again allows smooth curves to degenerate to nodal curves.

The problem of extending the theory of stable maps to define Gromov–Witten invariants for *stacks* has been an organizing question in the subject for some time. There is a very nice answer for smooth and proper Deligne–Mumford (DM) stacks, which involves allowing twisted curves – nodal curves with μ_r orbifold structure allowed at the nodes and marked points [AV02]. There is also a further generalization to tame algebraic stacks in positive characteristic [AOV11].

For non-orbifold quotients, and specifically for quotients X/G of affine schemes by reductive groups, the most developed theory is the theory of quasi-maps [CFKM14]. The theory applies to a Neron-Severi class $\ell \in \mathrm{NS}(X/G)_{\mathbb{Q}}$ for which G acts with finite stabilizers on the ℓ-semistable locus, $X^{\ell-\mathrm{ss}} \subset X$, in the sense of geometric invariant theory (GIT). For such an ℓ, the space of ℓ-stable quasi-maps parameterizes maps from nodal curves to X/G that land in $X^{\ell-\mathrm{ss}}$ away from finitely many base points and satisfy a certain stability condition. For a sufficiently large positive scaling of ℓ, it agrees with the space of stable maps to the GIT quotient orbifold $X^{\ell-\mathrm{ss}}/G$. Studying wall-crossing, i.e., how these moduli spaces depend on ℓ, has been significantly complicated by the breakdown of the theory for ℓ such that points of $X^{\ell-\mathrm{ss}}$ have positive dimensional stabilizers.

Naively speaking, the ideal answer to this circle of questions would be to identify some very large open substack of the stack $\mathcal{M}^{\mathrm{pre}}(X/G)$ of nodal curves with a map to X/G, and to show that different choices of stability parameters determine a Θ-stratification [HL22] of this large stack whose semistable locus admits a proper good moduli space, including the spaces of quasi-maps as special cases. Then one could use this stratification to study wall-crossing. However, this structure has not been identified even for the target BG, despite the question receiving attention from many perspectives, and progress for $G = \mathrm{GL}_N$ [Gie84, Cap94, NS99, Sch04, Kau10, HLH25].

This paper revisits the foundational assumption that one should focus on open substacks of $\mathcal{M}^{\mathrm{pre}}(X/G)$. Perhaps instead of plain nodal (twisted) curves, one should consider nodal curves along with some additional structure, which is vacuous when the curve is stable and which has good moduli-theoretic properties. Recall that a twisted curve is called *semistable* if the map to its stabilization contracts only chains of rational curves, called rational bridges, over nodes.

Definition 1.0.1 (Anchored curves)**.** The stack of anchored curves $\mathcal{A}nc_{g,n}$ sends a k-scheme T to the groupoid of tuples $(\mathcal{C}, \Sigma_\bullet, \mathcal{L}, s : \mathcal{L} \to \mathcal{O}_{\mathcal{C}})$, where

(1) $(\mathcal{C}, \Sigma_\bullet) \to T$ is a family of n-marked semistable twisted curves of genus $g \geq 2$ as in Definition 2.1.1, with coarse space $(C, \sigma_\bullet) \to T$,

(2) $\mathcal{L}$ is an invertible sheaf on $\mathcal{C}$, and

(3) $s : \mathcal{L} \to \mathcal{O}_{\mathcal{C}}$ is a cosection.

In addition, this data is required to satisfy the following conditions:

(a) The stabilizers of $\mathcal{C}$ act faithfully on the fibers of $\mathcal{L}$, i.e., $\mathcal{C} \to B\mathbb{G}_m$ is representable;

(b) $\mathcal{L}$ is ample relative to the stabilization morphism $(\mathcal{C}, \Sigma_\bullet) \to (C^{\mathrm{st}}, \sigma_\bullet^{\mathrm{st}})$, meaning that Zariski locally on T, there exists some positive integer $N > 0$ such that $\mathcal{L}^{\otimes N} := L$ descends to a line bundle L on C that is ample relative to C^{st};

(c) For all $t \in T$ the base-change $s_t : \mathcal{L}_t \to \mathcal{O}_{\mathcal{C}_t}$ is an isomorphism at every generic point of C_t^{st}, thought of as a point in $\mathcal{C}_t$.

We will let $\mathcal{X}_{g,n}$ denote the stack whose T-points consist of only the data (1) and (2) satisfying conditions (a) and (b). We will use the notation $\mathcal{A}nc_g := \mathcal{A}nc_{g,0}$.

We define the *degree* of a line bundle on a twisted curve $\mathcal{C}$ in terms of the Euler characteristic $\chi(-)$,

$$\deg(\mathcal{L}) := \frac{1}{N}\left(\chi(\mathcal{L}^{\otimes N}) - \chi(\mathcal{O}_{\mathcal{C}})\right),$$

where N is any common multiple of the order of all automorphism groups of $\mathcal{C}$. For such an N, $\mathcal{L}^{\otimes N}$ descends to the coarse moduli space of $\mathcal{C}$, and $\chi(\mathcal{L}^{\otimes N}) - \chi(\mathcal{O}_{\mathcal{C}})$ is the usual degree of the descended bundle. Note that $\deg(\mathcal{L})$ is a locally constant function on $\mathcal{A}nc_g$. We say that an anchored curve $(\mathcal{C}, \mathcal{L}, s)$ has degree 0 if $\mathcal{L}$ has degree 0.

If $(\mathcal{C}, \mathcal{L}, s)$ is a point of $\mathcal{A}nc_g$ such that the coarse moduli space is stable, then condition (c) above implies that the degree of $\mathcal{L}$ is non-positive on every component. Therefore, if $\deg(\mathcal{L}) = 0$, it must be that $s : \mathcal{L} \to \mathcal{O}_{\mathcal{C}}$ is an isomorphism, and then condition (a) implies that the automorphism groups of $\mathcal{C}$ are trivial. So, the additional data of $\mathcal{L}$, s, and possible orbifold structure are only visible on non-stable curves.

Definition 1.0.1 is inspired by the beyond GIT program [HL22, AHLH23, GFHZ24], which reduces the construction of a good moduli space to the conditions of Θ- and S-completeness. These conditions involve certain regular 2-dimensional schemes Y with a $\mathbb{G}_m$-action and a codimension-2 fixed point $\mathfrak{o} \in Y$, where the condition is that a morphism $(Y \setminus \mathfrak{o})/\mathbb{G}_m \to \mathcal{X}$ extends uniquely to $Y/\mathbb{G}_m$. The issue with $\mathcal{X} = \mathcal{M}^{\mathrm{pre}}(X/G)$ is that even if an extension exists, it might not be unique.

This is really a pathology with the stack of nodal curves, though. We will see many examples below of families of nodal curves over $Y \setminus \mathfrak{o}$ that extend non-uniquely to families over Y. This non-uniqueness is closely related to the failure of the diagonal to be affine (see Remark 2.2.2), which is in turn related to the lack of a canonical ample bundle on a nodal curve. Our Definition 1.0.1 addresses this pathology with the stack of all nodal curves.

Theorem 1.0.2 (= Theorem 2.2.7)**.** *The stacks $\mathcal{A}nc_{g,n}$ and $\mathcal{X}_{g,n}$ have affine diagonal, and they are Θ- and S-complete.*

This result inspires the following:

Proposal 1.0.3. Gromov–Witten style moduli problems for the stacks X/G, with X affine and G reductive, should be constructed relative to the stack of degree 0 anchored curves.

In a companion paper to this one [HLHIS26], we will realize this proposal when $X/G = B\,\mathrm{SL}_N$. We enlarge the stack of SL_N bundles on the universal curve over $\mathcal{M}_g$ by considering the stack of degree 0 anchored curves along with an SL_N bundle satisfying certain open conditions. The enlargement has several nice properties, including a Θ-stratification whose semistable locus admits a proper and flat good moduli space over $\overline{M}_g$.

In the second part of the paper, we study a smaller open substack $\mathrm{P}\overline{\mathcal{M}}_g \subset \mathcal{A}nc_g$. The motivation is that for a degree 0 point $(\mathcal{C}, \mathcal{L}, s)$ of $\mathcal{A}nc_g$, if the degree of $\mathcal{L}$ on the rational bridges of $\mathcal{C}$ is large, then the Cartier divisor $s^\vee : \mathcal{O}_{\mathcal{C}} \to \mathcal{L}^\vee$ on the stable components of $\mathcal{C}$ can deform inside the corresponding fiber over the stack $\mathfrak{M}_g^{tw}$ of twisted curves. On the other hand, if $(\mathcal{C}, \mathcal{L}, s)$ is a point of $\mathrm{P}\overline{\mathcal{M}}_g$, then $s^\vee$ is forced to vanish only on the rational bridges Lemma 3.1.1, and the divisor on the stable components of $\mathcal{C}$ is rigid in the fiber over $\mathfrak{M}_g^{tw}$.

By an *orientation* of a rational bridge $\mathcal{R} \subset \mathcal{C}$ in a twisted curve, we mean a choice of labels (p_-, p_+) for the two points on the stable components of $\mathcal{C}$ where $\mathcal{R}$ meets the rest of the curve. Given an orientation of $\mathcal{R}$, we let $\mathrm{ord}_\pm(\mathcal{R})$ denote the order of vanishing of the section $s^\vee$ at $p_\pm$ on the stable components, divided by the order of the stabilizer at that point.

Definition 1.0.4 (Relatively polarized curves)**.** The substack $\mathrm{P}\overline{\mathcal{M}}_g \subset \mathcal{A}nc_g$ of *relatively polarized curves* consists of T-families $(\mathcal{C}, \mathcal{L}, s)$ of degree 0 anchored curves such that for every geometric point $t \to T$, and for every rational bridge $\mathcal{R} \subset \mathcal{C}_t$, with some choice of orientation, one has

$$\mathrm{ord}_\pm(\mathcal{R}) \leq 1 \quad \text{and} \quad \mathrm{ord}_+(\mathcal{R}) + \mathrm{ord}_-(\mathcal{R}) = \deg(\mathcal{L}|_{\mathcal{R}}).$$

This notion fell out of our work [HLHIS26] on the stack of SL_N bundles on degree 0 anchored curves, when we noticed that the stack we study there lives over the smaller substack $\mathrm{P}\overline{\mathcal{M}}_g \subset \mathcal{A}nc_g$. We will discuss the connection more in [HLHIS26]

In this paper, we focus on a separate interesting phenomenon, first identified in [ABB+23]: $\mathrm{P}\overline{\mathcal{M}}_g$ admits an exhaustive ascending family of open substacks that all have the same good moduli space, $\overline{M}_g$. These substacks are parameterized by the following datum.

Hypothesis 1.0.5. Let $\mathcal{B}$ be a collection of finite subsets

$$\mathcal{B} \subset \{S \subset [-1,1] \cap \mathbb{Q} \mid |S| < \infty \text{ and } \max(S) > 0 > \min(S)\}$$

that satisfies the following conditions:

(1) $S \in \mathcal{B} \Rightarrow -S \in \mathcal{B}$

(2) $S, T \in \mathcal{B} \Rightarrow S \cup T \in \mathcal{B}$

(3) If $S \in \mathcal{B}$ and $T \subset S$ with $\max(T) > 0 > \min(T)$, then $T \in \mathcal{B}$.

To any oriented rational bridge $\mathcal{R}$ in a degree 0 anchored curve, in Definition 3.1.2 we associate a finite subset $\mathrm{type}(\mathcal{R}) \subset \mathbb{Q} \cap [-1,1]$ with $\max(\mathrm{type}(\mathcal{R})) > 0 > \min(\mathrm{type}(\mathcal{R}))$. See also Lemma 3.1.4(3) for an explicit description of the relation between $\mathrm{type}(\mathcal{R})$ and the isomorphism class of the pair $(\mathcal{R}, \mathcal{L}|_{\mathcal{R}})$.

Definition 1.0.6. For any $\mathcal{B}$ as in Hypothesis 1.0.5, let $\mathrm{P}\overline{\mathcal{M}}_g^{\mathcal{B}} \subset \mathrm{P}\overline{\mathcal{M}}_g$ denote the substack of points $(\mathcal{C}, \mathcal{L}, s)$ such that for every rational bridge $\mathcal{R} \subset \mathcal{C}$, $\mathrm{type}(\mathcal{R}) \in \mathcal{B}$.

When $\mathcal{B} = \emptyset$, then $\mathrm{P}\overline{\mathcal{M}}_g^{\mathcal{B}} = \overline{\mathcal{M}}_g$, and when $\mathcal{B}$ is the collection of all finite subsets, $\mathrm{P}\overline{\mathcal{M}}_g^{\mathcal{B}} = \mathrm{P}\overline{\mathcal{M}}_g$. Another way to produce such an $\mathcal{B}$ is to choose an arbitrary subset $X \subset [-1,1] \cap \mathbb{Q}$ such that $-X = X$, and let $\mathcal{B}$ be the collection of all finite subsets $S \subset X$ with $\max(S) > 0 > \min(S)$.

Theorem 1.0.7. *For any $\mathcal{B}$ as in Hypothesis 1.0.5, $\mathrm{P}\overline{\mathcal{M}}_g^{\mathcal{B}} \subset \mathcal{A}nc_g$ is an open substack that is Θ- and S-complete. If $\mathcal{B}$ is finite, then $\mathrm{P}\overline{\mathcal{M}}_g^{\mathcal{B}}$ is bounded and $\mathrm{P}\overline{\mathcal{M}}_g^{\mathcal{B}} \to \overline{M}_g$ is a good moduli space.*

Notation and conventions

We work over a field k of characteristic 0. Unless otherwise stated, all of our schemes and algebraic stacks are implicitly assumed to be locally Noetherian. We use the term DVR to refer to a discrete valuation ring. Given a local ring A, we will denote by A^{sh} its strict henselization. For an algebraic space X equipped with an action of a group scheme G, we denote the quotient stack by X/G (as opposed to $[X/G]$).

Throughout the paper, we fix an integer $g \geq 2$, which will be the genus of the curves in our moduli problems.

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2 Anchored curves

In this section we define the stack $\mathcal{A}nc_{g,n}$ of anchored curves, see Definition 1.0.1. Roughly speaking, $\mathcal{A}nc_{g,n}$ parameterizes n-marked twisted semistable curves $\mathcal{C}$ of genus g equipped with a line bundle $\mathcal{L}$ that is ample relative to the stabilization and is trivialized generically on each irreducible component of $\mathcal{C}$ that is not contracted by the stabilization morphism. The main theorem of this section is Theorem 2.2.7, where we prove that the stack $\mathcal{A}nc_{g,n}$ satisfies the Θ-completeness and S-completeness conditions introduced in [AHLH23].

2.1 The Moduli of Anchored Curves

Preliminaries on Twisted Curves

We denote by $\mathfrak{M}_{g,n}$ the smooth stack of prestable marked curves of genus g with n marked points. For a marked prestable curve $(C, (\sigma_i)_{i=1}^n)$ defined over a field, we denote by ω_C the canonical line bundle of C. We call $\omega_C(\sum_{i=1}^n \sigma_i)$ the log canonical bundle. We consider two open substacks of $\mathfrak{M}_{g,n}$:

- The substack $\overline{\mathcal{M}}_{g,n}$ of stable curves; curves whose log canonical bundle is ample.
- The substack $\mathfrak{M}_{g,n}^{ss}$ of semistable curves; curves whose log canonical bundle is nef.

The stack of stable curves $\overline{\mathcal{M}}_{g,n}$ is known to be smooth, proper, Deligne–Mumford and irreducible. The open inclusion $\overline{\mathcal{M}}_{g,n} \hookrightarrow \mathfrak{M}_{g,n}$ admits a retract $\mathrm{st} : \mathfrak{M}_{g,n} \to \overline{\mathcal{M}}_{g,n}$ called the stabilization morphism, which sends a family of marked curves $(C, (\sigma_i)_{i=1}^n)$ to its stabilization $(C^{\mathrm{st}}, (\sigma_i^{\mathrm{st}})_{i=1}^n)$. The morphism $\mathrm{st} : \mathfrak{M}_{g,n} \to \overline{\mathcal{M}}_{g,n}$ of smooth stacks is syntomic by [Beh97, Prop. 3], and is of relative dimension 0 by standard deformation theory of nodal curves.

Given a prestable marked curve $(C, \sigma_\bullet) \in \mathfrak{M}_{g,n}$ defined over an algebraically closed field, recall that a maximal connected subcurve $R \subset C$ contracted by the stabilization morphism $(C, \sigma_\bullet) \to (C^{\mathrm{st}}, \sigma_\bullet^{\mathrm{st}})$ is called a rational bridge if it is an unmarked chain of rational curves such that the intersection $R \cap \overline{C \setminus R}$ consists of exactly two closed points of C (cf. [Sta25, Tag 0E3M]). We call these two points the end points of the bridge. On the other hand, such a maximal connected subcurve $R \subset C$ is called a rational tail if it is a chain of rational curves such that $R \cap \overline{C \setminus R}$ consits of one point, which we call the attaching point of the tail (cf. [Sta25, Tag 0E3G]). If $R \subset C$ is a rational tail, then there is a unique irreducible component $\Sigma \subset R$ such that $\omega_C|_\Sigma$ is negative; we call this the terminal component of the tail.

A marked curve $(C, \sigma_\bullet) \in \mathfrak{M}_{g,n}$ defined over an algebraically closed field is semistable if and only if any maximal connected reduced subcurve $R \subset C$ contracted by the stabilization morphism $C \to C^{\mathrm{st}}$ is either:

- a rational bridge (with no markings), or
- a rational tail that contains exactly one marking located in its terminal component.

Recall the following notion of a n-marked twisted curve as in [Ols07, Def. 1.2].

Definition 2.1.1. An n-marked twisted curve of genus g over a k-scheme T is a morphism $f : \mathcal{C} \to T$ together with a tuple of disjoint closed substacks $(\Sigma_i)_{i=1}^n$ of $\mathcal{C}$ such that:

(1) $\mathcal{C}$ is a proper, finitely presented, with connected one-dimensional geometric fibres. Its coarse moduli space $\mathcal{C} \to C$ is a nodal curve of genus g over T.

(2) Each Σ_i is an étale gerbe over T, and it's image in C is contained in the smooth locus of $C \to T$.

(3) If $\mathcal{U} \subset \mathcal{C}$ denotes the complement of the Σ_i and the singular locus of $\mathcal{C} \to T$, then $\mathcal{U} \to C$ is an open immersion.

(4) For every geometric point $\bar{t} \to T$ whose image lies on a marking, there is an integer $r \geq 1$ such that étale locally, the fibre $\mathcal{C}_{\bar{t}}$ near that marking is isomorphic to

$$\mathrm{Spec}(k(\bar{t})[z])/\mu_r,$$

where $\zeta \in \mu_r$ acts by $z \mapsto \zeta z$,

(5) For every geometric point $\bar{t} \to T$ whose image is a node, there is an integer $r \geq 1$ such that étale locally, $\mathcal{C}_{\bar{t}}$ near that node is isomorphic to

$$\mathrm{Spec}(k(\bar{t})[x, y]/(xy))/\mu_r,$$

where $\zeta \in \mu_r$ acts by $(x, y) \mapsto (\zeta x, \zeta^{-1} y)$.

We denote by $\mathfrak{M}_{g,n}^{tw}$ the smooth algebraic stack parameterizing families of n-marked twisted curves of genus g [Ols07, Thm. 1.10]. The open immersion $\mathfrak{M}_{g,n} \hookrightarrow \mathfrak{M}_{g,n}^{tw}$ admits a retract $c : \mathfrak{M}_{g,n}^{tw} \to \mathfrak{M}_{g,n}$ given by sending a family of marked twisted curves to its coarse space. It follows from the local description in [Ols07, §5] that the morphism c is syntomic of relative dimension 0. We denote by $\mathfrak{M}_{g,n}^{tw,ss} := c^{-1}(\mathfrak{M}_{g,n}^{ss}) \subset \mathfrak{M}_{g,n}^{tw}$ the open substack of marked twisted curves whose coarse space is semistable. We call families in $\mathfrak{M}_{g,n}^{tw,ss}$ marked twisted semistable curves. The stabilization morphism for twisted semistable curves is the composition $\mathrm{st} : \mathfrak{M}_{g,n}^{tw,ss} \xrightarrow{c} \mathfrak{M}_{g,n}^{ss} \xrightarrow{\mathrm{st}} \overline{\mathcal{M}}_{g,n}$, which is syntomic of relative dimension 0. Given a marked twisted semistable curve over an algebraically closed field $\mathcal{C}$, a reduced closed substack $\mathcal{R} \subset \mathcal{C}$ is called a rational bridge (resp. rational tail) if its image under the coarse space morphism $\mathcal{C} \to C$ is a rational bridge (resp. rational tail). We say that $\mathcal{C}$ has ℓ rational bridges (resp. tails) if its coarse space C does. Given a rational bridge $\mathcal{R} \subset \mathcal{C}$, its end points are defined to be the preimages of the end points of the corresponding rational bridge $R \subset C$ in the coarse space. Similarly, we define the attaching point of a rational tail $\mathcal{R} \subset \mathcal{C}$ as the preimage of the attaching point in the coarse space.

Remark 2.1.2. Given a twisted n-marked semistable curve $(\mathcal{C}, \Sigma_\bullet) \to S$ over a scheme S, we will usually denote by $\varphi : (\mathcal{C}, \Sigma_\bullet) \to (C, \sigma_\bullet)$ its coarse space and by $\pi : (C, \sigma_\bullet) \to (C^{\mathrm{st}}, \sigma_\bullet^{\mathrm{st}})$ the stabilization of the coarse space. If the number of markings is $n = 0$, then we have identifications of canonical bundles $\omega_{\mathcal{C}/S} \cong \varphi^*(\omega_{C/S}) = \varphi^*\pi^*(\omega_{C^{\mathrm{st}}/S})$, where the first equality follows from [Chi08, Prop. 2.5.1] and the second equality follows from the curve C being semistable. In particular, Grothendieck duality implies that for any given vector bundles $\mathcal{E}$ and $\mathcal{F}$ on C^{st} and $\mathcal{C}$ respectively, there is a canonical correspondence between morphisms $\mathcal{F} \to \varphi^*\pi^*(\mathcal{E})$ and morphisms $\pi_*\varphi_*(\mathcal{F}) \to \mathcal{E}$. We shall use this fact at several points in the paper.

The Moduli of Anchored Curves

Recall the stacks $\mathcal{A}nc_{g,n}$ and $\mathcal{X}_{g,n}$ from Definition 1.0.1.

Proposition 2.1.3. *$\mathcal{X}_{g,n}$ and $\mathcal{A}nc_{g,n}$ are algebraic stacks locally of finite type and with affine diagonal. The forgetful morphism*

$$\mathcal{A}nc_{g,n} \to \mathcal{X}_{g,n}, \quad (\mathcal{C}, \Sigma_\bullet, \mathcal{L}, s : \mathcal{L} \to \mathcal{O}_\mathcal{C}) \mapsto (\mathcal{C}, \Sigma_\bullet, \mathcal{L})$$

is quasiaffine and of finite type.

Proof. Let us first check that $\mathcal{X}_{g,n}$ is an algebraic stack locally of finite type over k. If we denote by $(\mathcal{C}_{univ}, \Sigma_\bullet) \to \mathrm{Map}_{\mathfrak{M}_g^{tw,ss}}$ the universal family, then the mapping stack $\mathrm{Map}_{\mathfrak{M}_g^{tw,ss}}(\mathcal{C}_{univ}, B\mathbb{G}_m)$ is algebraic and locally of finite type over k by [HR19, Thm. 1.2]. Note that $\mathrm{Map}_{\mathfrak{M}_g^{tw,ss}}(\mathcal{C}_{univ}, B\mathbb{G}_m)$ parameterizes pairs of a twisted semistable curve $(\mathcal{C}, \Sigma_\bullet)$ and a line bundle $\mathcal{L}$ on $\mathcal{C}$. The inclusion $\mathcal{X}_{g,n} \hookrightarrow \mathrm{Map}_{\mathfrak{M}_g^{tw,ss}}(\mathcal{C}_{univ}, B\mathbb{G}_m)$ is an open immersion, because of the following:

- The condition on $\mathcal{L}$ being ample relative to the stabilization is open.
- The condition on the representability of the morphism $f : \mathcal{C} \to B\mathbb{G}_m$ corresponding to $\mathcal{L}$ is open. Indeed, the relative inertia I_f is a closed substack of the inertia stack $I_\mathcal{C}$, which is finite and representable over $\mathcal{C}$. The identity section $\mathcal{C} \to I_f$ induces surjection $\mathcal{O}_{I_f} \twoheadrightarrow \mathcal{O}_\mathcal{C}$ of coherent sheaves on $\mathcal{C}$. Let $\mathcal{K}$ denote the kernel of this surjection; it is a coherent sheaf on $\mathcal{C}$ whose formation commutes with base-change by the flatness of $\mathcal{O}_\mathcal{C}$. Then the locus of geometric points of T where the morphism f is representable is the open complement of the image of the support $\mathrm{Supp}(\mathcal{K}) \subset \mathcal{C}$ under the proper morphism $\mathcal{C} \to T$.

In particular, $\mathcal{X}_{g,n}$ is algebraic and locally of finite type over k.

Next, we show that $\mathcal{X}_{g,n}$ has affine diagonal. It is enough to show that any quasi-compact open substack $\mathcal{U} \subset \mathcal{X}_{g,n}$ has affine diagonal. By quasi-compactness, there exists some number N such that for all $(\mathcal{C}, \Sigma_\bullet, \mathcal{L})$ families in $\mathcal{U}$, the line bundle $\mathcal{L}^{\otimes N}$ descends to the coarse space $(C, \sigma_\bullet)$ and $\mathcal{L}^{\otimes N} \otimes \omega_C(\sum_{i=1}^n \sigma_i)^{\otimes N^2}$ defines an ample line bundle on C. Hence, we have a well-defined assignment

$$F : \mathcal{U} \to \mathfrak{M}_{g,n}^{ss,pol}, \quad (\mathcal{C}, \Sigma_\bullet, \mathcal{L}) \mapsto (C, \sigma_\bullet, \mathcal{L}^{\otimes N} \otimes \omega_C(\sum_{i=1}^n \sigma_i)^{\otimes N^2})$$

to the stack $\mathfrak{M}_{g,n}^{ss,pol}$ of semistable curves equipped with an ample line bundle. Since $\mathfrak{M}_{g,n}^{ss,pol}$ has affine diagonal [HLH25, Prop. 2.3], it suffices to show that F has affine relative diagonal. Let $(C_{univ}, \sigma_\bullet, \mathcal{M}) \to \mathfrak{M}_{g,n}^{ss,pol}$ denote the universal polarized curve. We denote by $\mathcal{Z} \to C_{univ}$ the Nth-root stack of the line bundle $\mathcal{M} \otimes \omega_{C_{univ}/\mathfrak{M}^{ss,pol}}(\sum_{i=1}^n \sigma_i)^{\otimes(-N^2)}$. Then $\mathcal{Z} \to C_{univ} \to \mathfrak{M}_{g,n}^{ss,pol}$ is a relatively DM morphism which is locally projective. Hence we may define the stack $\mathcal{K}_{g,n}(\mathcal{Z}/\mathfrak{M}_{g,n}^{ss,pol})$ of n-marked twisted stable maps relative to $\mathfrak{M}_{g,n}^{ss,pol}$ as in [AV02, §8.3]. By the definition of $\mathcal{X}_{g,n}$, there is an inclusion $\mathcal{X}_{g,n} \hookrightarrow \mathcal{K}_g(\mathcal{Z}/\mathfrak{M}^{ss,pol})$ which realizes $\mathcal{X}_{g,n}$ as the locally closed substack of T-families of twisted stable maps $(\mathcal{C}, \Sigma_\bullet) \to \mathcal{Z}_T$ such that the morphism of T-flat families $(\mathcal{C}, \Sigma_\bullet) \to ((C_{univ})_T, \sigma_\bullet|_T)$ is a coarse space morphism compatible with the markings. In particular, since the stack of twisted stable maps $\mathcal{K}_{g,n}(\mathcal{Z}/\mathfrak{M}_{g,n}^{ss,pol})$ is separated over $\mathfrak{M}_{g,n}^{ss,pol}$ [AV02, Thm.1.4.1+§8.3], the same holds for $\mathcal{X}_{g,n}$.

We are left to show that the forgetful morphism $\mathcal{A}nc_{g,n} \to \mathcal{X}_{g,n}$ is quasiaffine and of finite type. Let $\mathcal{Y} \to \mathcal{X}_{g,n}$ denote the stack parameterizing triples $(\mathcal{C}, \Sigma_\bullet, \mathcal{L}, s :$

$\mathcal{L} \to \mathcal{O}_{\mathcal{C}}$), where $(\mathcal{C}, \Sigma_\bullet, \mathcal{L})$ is a point in $\mathcal{X}_{g,n}$ and s is a cosection. By [Sta25, Tag 08K6], it follows that the forgetful morphism $\mathfrak{Y} \to \mathcal{X}_{g,n}$ is affine and of finite type. Furthermore, the inclusion $\mathcal{A}nc_{g,n} \hookrightarrow \mathfrak{Y}$ is an open immersion because of the condition of s being nonvanishing at every generic pont of the stabilization is open. Indeed, let $\varphi : \mathcal{C} \to C^{\mathrm{st}}$ denote the stabilization of the coarse space, and consider the closed image $Z \subset C^{\mathrm{st}}$ of the vanishing locus $V(s) \subset \mathcal{C}$ under the proper morphism φ. By upper semicontinuity of fiber dimension, there is a closed subset $Z_1 \subset Z$ of points where the dimension of the fiber of $Z \to T$ is at least 1. Then the locus of geometric points $\bar{t} \to T$ where the cosection satisfies this generic vanishing condition is the open complement of the image of the proper morphism $Z_1 \to T$. We conclude that the composition $\mathcal{A}nc_{g,n} \hookrightarrow \mathfrak{Y} \to \mathcal{X}_{g,n}$ is quasiaffine and of finite type. □

Definition 2.1.4. We define $\mathcal{X}^{\mathrm{rep}}_{g,n} \subset \mathcal{X}_{g,n}$ (resp. $\mathcal{A}nc^{\mathrm{rep}}_{g,n} \subset \mathcal{A}nc_{g,n}$) to be the substack that for any k-scheme T parameterizes T-points $(\mathcal{C}, \Sigma_\bullet, \mathcal{L})$ of $\mathcal{X}_{g,n}$ (resp. $(\mathcal{C}, \Sigma_\bullet, \mathcal{L}, s)$ of $\mathcal{A}nc_{g,n}$) such that for all $t \in T$ the twisted curve $\mathcal{C}_t$ is a scheme.

Proposition 2.1.5. *The inclusions $\mathcal{X}^{\mathrm{rep}}_{g,n} \hookrightarrow \mathcal{X}_{g,n}$ and $\mathcal{A}nc^{\mathrm{rep}}_{g,n} \hookrightarrow \mathcal{A}nc_{g,n}$ are open immersions, and they fit into a Cartesian diagram*

$$\begin{array}{ccc} \mathcal{A}nc^{\mathrm{rep}}_{g,n} & \longrightarrow & \mathcal{A}nc_{g,n} \\ \downarrow & & \downarrow \\ \mathcal{X}^{\mathrm{rep}}_{g,n} & \longrightarrow & \mathcal{X}_{g,n}. \end{array}$$

Proof. Consider the forgetful morphisms $\mathcal{A}nc_{g,n} \to \mathcal{X}_{g,n} \to \mathfrak{M}^{tw}_{g,n}$ to the stack of twisted marked curves. By definition, the substacks $\mathcal{X}^{\mathrm{rep}}_{g,n}$ and $\mathcal{A}nc^{\mathrm{rep}}_{g,n}$ are the preimages of the substack $\mathfrak{M}^{tw,\mathrm{rep}}_{g,n} \subset \mathfrak{M}^{tw}_{g,n}$ parameterizing T-families $(\mathcal{C}, \Sigma_\bullet)$ such that for all $t \in T$ the twisted curve $\mathcal{C}_t$ is a scheme. The result follows because $\mathfrak{M}^{tw,\mathrm{rep}}_{g,n} \hookrightarrow \mathfrak{M}^{tw}_{g,n}$ is known to be an open immersion [Ols07, Cor. 1.12]. □

2.2 Codimension Two Filling Conditions

The goal of this section is to prove Θ-completeness and S-completeness of the stacks $\mathcal{X}_{g,n}$ and $\mathcal{A}nc_{g,n}$. We recall both of these codimension-two filling conditions below. Throughout, we use R to denote a DVR with uniformizer π and fraction field K.

Definition 2.2.1. Let $\mathcal{X}$ be an algebraic stack over k. We consider commutative diagrams of the form

$$\begin{array}{ccc} \mathcal{Y}^\circ & \longrightarrow & \mathcal{X} \\ \downarrow & \nearrow & \\ \mathcal{Y} & & \end{array} \tag{1}$$

where $\mathcal{Y}$ is an algebraic stack with closed point $\mathfrak{o} \in \mathcal{Y}$, and $\mathcal{Y}^\circ := \mathcal{Y} \setminus \mathfrak{o}$. For two different pairs $(\mathcal{Y}, \mathfrak{o})$, we say that the associated filling condition holds if for every such diagram (1) of solid arrows there exists a unique dotted arrow making the diagram commute.

(1) Define $\overline{\mathrm{ST}}_R := \mathrm{Spec}(R[s,t]/(st - \pi))/\mathbb{G}_m$ where $\mathbb{G}_m$ acts by giving s weight 1 and t weight -1. Let $\mathfrak{o}$ be the closed point corresponding to the ideal (s,t). We say that $\mathcal{X}$ is *S-complete* if the filling condition holds for $(\mathcal{Y} = \overline{\mathrm{ST}}_R, \mathfrak{o})$ for every DVR R.

(2) Define $\Theta_R := \mathrm{Spec}(R[t])/\mathbb{G}_m$, where $\mathbb{G}_m$ acts by giving t weight -1. Let $\mathfrak{o}$ be the closed point corresponding to the ideal (π, t). We say that $\mathcal{X}$ is Θ-*complete* if the filling condition holds for $(\mathcal{Y} = \Theta_R, \mathfrak{o})$ for every DVR R.

Remark 2.2.2. If the stack $\mathcal{X}$ has affine diagonal, then the uniqueness of the filling in (1) is automatic, and to check the existence we may replace R with an arbitrary extension of DVR's $R \subset R'$ [AHLH23, Prop. 3.17, Prop. 3.41]. We use this fact freely in our proofs of Θ-completeness and S-completeness.

Notation 2.2.3. In the setting of Definition 2.2.1, each of the test stacks $\mathcal{Y}$ arises as a quotient $\mathcal{Y} = Y/\mathbb{G}_m$, where for $\mathcal{Y} = \overline{\mathrm{ST}}_R$, we set $Y := \mathrm{Spec}(R[s,t]/(st - \pi))$ and for $\mathcal{Y} = \Theta_R$, we set $Y := \mathrm{Spec}(R[t])$. Abusing notation, we also write $\mathfrak{o} \in Y$ for the closed point lying over $\mathfrak{o} \in \mathcal{Y}$ cut out by the corresponding ideal (s,t) in the S-completeness case and (π, t) in the Θ-completeness case. We set $Y^\circ := Y \setminus \mathfrak{o}$, so that $\mathcal{Y}^\circ = Y^\circ/\mathbb{G}_m$.

Some reduction results for filling conditions

Lemma 2.2.4. *The forgetful morphism $\mathcal{A}nc_{g,n} \to \mathcal{X}_{g,n}$ is Θ-complete and S-complete. In other words, given any DVR R and letting $\mathcal{Y}$ be as in Notation 2.2.3, for any diagram of solid arrows as below*

$$\begin{array}{ccc} \mathcal{Y}^\circ & \longrightarrow & \mathcal{A}nc_{g,n} \\ \downarrow & \nearrow & \downarrow \\ \mathcal{Y} & \longrightarrow & \mathcal{X}_{g,n} \end{array} \tag{2}$$

there exists a unique dotted arrow making the diagram commute. The same holds for the morphism $\mathcal{A}nc^{\mathrm{rep}}_{g,n} \to \mathcal{X}^{\mathrm{rep}}_{g,n}$.

Proof. In view of the Cartesian diagram in Proposition 2.1.5, it suffices to prove the result for the morphism $\mathcal{A}nc_{g,n} \to \mathcal{X}_{g,n}$. The uniqueness of the dotted arrow in (2) follows from the fact that $\mathcal{A}nc_{g,n}$ has affine diagonal (see Remark 2.2.2). We just need to show the existence. The morphism $\mathcal{Y}^\circ \to \mathcal{A}nc_{g,n}$ corresponds to a family $(\mathcal{C}^\circ, \Sigma^\circ_\bullet, \mathcal{L}^\circ, s^\circ)$ over $\mathcal{Y}^\circ$. By assumption, the triple $(\mathcal{C}^\circ, \Sigma^\circ_\bullet, \mathcal{L}^\circ)$ extends to a family $(\mathcal{C}, \Sigma_\bullet, \mathcal{L})$ over $\mathcal{Y}$. Furthermore, by Hartogs's theorem, the cosection $s^\circ : \mathcal{L}^\circ \to \mathcal{O}_{\mathcal{C}^\circ}$ extends uniquely to $s : \mathcal{L} \to \mathcal{O}_{\mathcal{C}}$. To conclude, we need to show that the tuple $(\mathcal{C}, \Sigma_\bullet, \mathcal{L}, s)$ satisfies condition (c) in Definition 1.0.1. Consider the stabilization $(C^{\mathrm{st}}, \sigma^{\mathrm{st}}_\bullet) \to \mathcal{Y}$ of the marked twisted curve $(\mathcal{C}, \Sigma_\bullet)$. Since $\overline{\mathcal{M}}_{g,n}$ is separated and has finite inertia, the morphism $(C^{\mathrm{st}}, \sigma^{\mathrm{st}}_\bullet) : \mathcal{Y} \to \overline{\mathcal{M}}_{g,n}$ factors through $\mathcal{Y} \to \mathrm{Spec}(R)$ [AHLH23, Lem. 3.6, Lem. 3.45]. In other words, there is a family of marked stable curves $(\overline{C}^{\mathrm{st}}, \overline{\sigma}^{\mathrm{st}}_\bullet)$ over $\mathrm{Spec}(R)$ such that $(C^{\mathrm{st}}, \sigma^{\mathrm{st}}_\bullet)$ is isomorphic to the base-change $(\overline{C}^{\mathrm{st}}_{\mathcal{Y}}, (\overline{\sigma}^{\mathrm{st}}_\bullet)_{\mathcal{Y}})$. To conclude we show that the following holds.

Claim. There is an open subscheme $U \subset \overline{C}^{\mathrm{st}}$ which is fiberwise dense over $\mathrm{Spec}(R)$ and satisfies the following:

(a) The morphism $\mathcal{C} \to C^{\mathrm{st}} \cong \overline{C}^{\mathrm{st}}_{\mathcal{Y}}$ is an isomorphism over the open $U_{\mathcal{Y}} \subset \overline{C}^{\mathrm{st}}_{\mathcal{Y}}$.

(b) The restriction of $s^\circ : \mathcal{L}^\circ \to \mathcal{O}_{\mathcal{C}_{\mathcal{Y}^\circ}}$ to $U_{\mathcal{Y}^\circ} \subset \mathcal{C}_{\mathcal{Y}^\circ}$ is an isomorphism.

If the claim holds, then Hartogs's theorem guarantees that s is an isomorphism when restricted to $U_{\mathcal{Y}}$, which is fiberwise dense inside $C^{\mathrm{st}} \cong \overline{C}^{\mathrm{st}}_{\mathcal{Y}}$, thus concluding the proof.

We are left to show the claim. Set $V \subset \overline{C}^{\mathrm{st}}$ to be the complement $\overline{C}^{\mathrm{st},sm} \setminus \bigcup_{i=1}^n \overline{\sigma}^{\mathrm{st}}_i$ of the markings inside the smooth locus $\overline{C}^{\mathrm{st},sm} \subset \overline{C}^{\mathrm{st}}$ of $\overline{C}^{\mathrm{st}} \to \mathrm{Spec}(R)$. Note that

the morphism $\mathcal{C} \to \overline{C}^{\mathrm{st}}_{\mathcal{Y}}$ is an isomorphism when restricted to $V_{\mathcal{Y}}$. Let s°_V denote the restriction of $s^\circ : \mathcal{L}^\circ \to \mathcal{O}_{\mathcal{C}^\circ}$ to $V_{\mathcal{Y}^\circ} \subset \mathcal{C}^\circ$. Let $W \subset V_{\mathcal{Y}^\circ}$ denote the open subset where s°_V is an isomorphism. Since s° is the cosection corresponding to the original morphism $\mathcal{Y}^\circ \to \mathcal{A}nc_{g,n}$, the open subset $W \subset \overline{C}^{\mathrm{st}}_{\mathcal{Y}^\circ} \cong C^{\mathrm{st}}_{\mathcal{Y}^\circ}$ has dense $\mathcal{Y}^\circ$-fibers. Let H be the reduced closed complement of $W \subset \overline{C}^{\mathrm{st}}_{\mathcal{Y}^\circ}$, denote by $\pi : \mathcal{Y}^\circ \to \mathrm{Spec}(R)$ the structure morphism, and let $Z \subset \overline{C}^{\mathrm{st}}$ be the schematic closure of H under the base-change $\pi_{\overline{C}^{\mathrm{st}}} : \overline{C}^{\mathrm{st}}_{\mathcal{Y}^\circ} \to \overline{C}^{\mathrm{st}}$. We set U to be the open complement of $Z \subset \overline{C}^{\mathrm{st}}$. By construction U satisfies conditions (a) and (b) of the claim; we are only left to prove that the $\mathrm{Spec}(R)$-fibers of $U \subset \overline{C}^{\mathrm{st}}$ are dense. Equivalently, we need to show that $Z \to \mathrm{Spec}(R)$ has finite fibers. To do this, we distinguish the cases of $\mathcal{Y} = \Theta_R$ and $\mathcal{Y} = \overline{ST}_R$.

If $\mathcal{Y} = \overline{ST}_R$, then the scheme $\mathcal{Y}^\circ$ is the union of two copies of $\mathrm{Spec}(R)$ glued at their generic points. In particular, the morphism $\mathcal{Y}^\circ \to \mathrm{Spec}(R)$ is quasifinite and universally closed. It follows that, set-theoretically, $Z \subset \overline{C}^{\mathrm{st}}$ is the naive image of H under $\pi_{\overline{C}^{\mathrm{st}}}$. Since the fibers of $H \to \mathcal{Y}^\circ$ are finite, so are the fibers of $H \to \mathcal{Y}^\circ \to \mathrm{Spec}(R)$, and hence the same holds for the fibers of the image $Z \subset \overline{C}^{\mathrm{st}}$.

Suppose now that $\mathcal{Y} = \Theta_R$. Let η denote the generic point of $\mathrm{Spec}(R)$. Note that $\mathcal{Y}^\circ$ is obtained by gluing $\mathcal{Y}^\circ_{t\neq 0} = \mathrm{Spec}(R)$ and Θ_η along their open points. Since the morphism $\pi_{\overline{C}^{\mathrm{st}}}$ restricts to an isomorphism over the open $\mathcal{Y}^\circ_{t\neq 0}$, it follows that the schematic image of $H|_{\mathcal{Y}^\circ_{t\neq 0}}$ has finite fibers. It is enough to show that the schematic image of $H|_{\Theta_\eta}$ under the morphism $\pi_{\overline{C}^{\mathrm{st}}} : \overline{C}^{\mathrm{st}}_{\Theta_\eta} \to \overline{C}^{\mathrm{st}}_{\eta} \hookrightarrow \overline{C}^{\mathrm{st}}$ is finite over $\mathrm{Spec}(R)$. This coincides with the closure of the (schematic) image of H_{Θ_η} under the closed projection morphism $\overline{C}^{\mathrm{st}}_{\Theta_\eta} \to \overline{C}^{\mathrm{st}}_{\eta}$; it suffices to show that the latter one is finite over η. For this we note that, over the open $V_{\Theta_\eta} \subset \overline{C}^{\mathrm{st}}_{\Theta_\eta}$, we have by assumption that the line bundle $\mathcal{L}^\circ|_{V_{\Theta_\eta}}$ is isomorphic to $\mathcal{O}_{V_{\Theta_\eta}}$ over a Θ_η-fiberwise dense open. In particular, the stabilizers of the stack V_{Θ_η} act trivially on the fibers of the line bundle $\mathcal{L}^\circ|_{V_{\Theta_\eta}}$, and therefore $\mathcal{L}^\circ|_{V_{\Theta_\eta}}$ is isomorphic to the pullback of a line bundle L on V_η under the good moduli space morphism $V_{\Theta_\eta} \to V_\eta$ [Alp13, Thm. 10.3]. By the projection formula applied to $V_{\Theta_\eta} \to V_\eta$, it follows that the restriction of the cosection $s^\circ|_{V_{\Theta_\eta}} : \mathcal{L}^\circ|_{V_{\Theta_\eta}} \to \mathcal{O}_{V_{\Theta_\eta}}$ is in fact the pullback of a cosection $\overline{s} : L \to \mathcal{O}_{V_\eta}$. Let $\overline{H} \subset V_\eta$ denote the finite η-scheme where $\overline{s}$ vanishes. Then H_{Θ_η} is the preimage under $\overline{C}^{\mathrm{st}}_{\Theta_\eta} \to \overline{C}^{\mathrm{st}}_{\eta}$ of the union $\overline{H} \cup (\overline{C}^{\mathrm{st}}_{\eta} \setminus V_\eta)$ of finite η-schemes. Therefore the image of H_{Θ_η} is the finite η-scheme $\overline{H} \cup (\overline{C}^{\mathrm{st}}_{\eta} \setminus V_\eta)$. □

Lemma 2.2.5. *Let R be a DVR, and let $\mathcal{Y}$ be either $\overline{ST}_R$ or Θ_R as in Notation 2.2.3. Let $\mathcal{Y}^\circ \to \mathcal{X}_{g,n}$ be a morphism corresponding to a tuple $(\mathcal{C}^\circ, \Sigma^\circ_\bullet, \mathcal{L}^\circ)$, and let $(C^\circ, \sigma^\circ_\bullet)$ denote the relative coarse space of $\mathcal{C}^\circ$. Let $N > 0$ be an integer such that $(\mathcal{L}^\circ)^{\otimes N}$ descends to a line bundle L° on C°, and consider the corresponding morphism $(C^\circ, \sigma^\circ_\bullet, L^\circ) : \mathcal{Y}^\circ \to \mathcal{X}^{\mathrm{rep}}_{g,n}$. If there is an extension to a morphism $(C, \sigma_\bullet, L) : \mathcal{Y} \to \mathcal{X}^{\mathrm{rep}}_{g,n}$, then there exists an extension of the original family $(\mathcal{C}^\circ, \Sigma^\circ_\bullet, \mathcal{L}^\circ)$ to a morphism $\mathcal{Y} \to \mathcal{X}_{g,n}$.*

Proof. Let us denote by $\mathcal{C}^\circ \to \overline{\mathcal{C}}^\circ \to \mathcal{Y}^\circ$ the partial relative coarse space where we only remove the fiberwise stackyness along the markings, so that $\mathcal{C}^\circ \to \overline{\mathcal{C}}^\circ$ is an isomorphism away from $\bigcup_i \Sigma^\circ_i$. There exists some twist $\mathcal{L}^\circ(\sum_i a_i \Sigma^\circ_i)$ that descends to a line bundle $\mathcal{M}^\circ$ on $\overline{\mathcal{C}}^\circ$. Note that $(\mathcal{L}^\circ(\sum_i a_i \Sigma^\circ_i))^{\otimes N}$ (or, equivalently, $(\mathcal{M}^\circ)^{\otimes N}$) descends to a twist $M^\circ := L^\circ(\sum_i b_i \sigma^\circ_i)$ of L°.

Set $M := L(\sum_i b_i \sigma_i)$, and let $\mathcal{R} \to C$ denote the Nth-root stack of the line bundle M. The morphism $\mathcal{R} \to \mathcal{Y}$ is proper and relatively DM with relative coarse space

C, which is projective smooth-locally over $\mathcal{Y}$. Let $\mathcal{K}_g(\mathcal{R}/\mathcal{Y}) \to \mathcal{Y}$ denote relative stack of twisted stable maps, which sends a scheme $T \to \mathcal{Y}$ to the groupoid of morphisms $f : \mathcal{S} \to \mathcal{R}_T$, where $\mathcal{S} \to T$ is a T-family of twisted curves of genus g and f is a respresentable morphism of T-stacks that induces a T-family of stable maps $\overline{f} : S \to C_T$ at the level of coarse spaces. Note that the stack $\mathcal{K}_g(\mathcal{R}/\mathcal{Y})$ is algebraic, relative DM and locally of finite type over $\mathcal{Y}$, and the assignment

$$\Psi : \mathcal{K}_g(\mathcal{R}/\mathcal{Y}) \to \mathcal{K}_g(C/\mathcal{Y}), \quad [f : \mathcal{S} \to \mathcal{R}_T] \mapsto [\overline{f} : S \to C_T]$$

is proper and quasifinite. Indeed, these properties can be checked smooth-locally over $\mathcal{Y}$, and when we replace $\mathcal{Y}$ with an atlas they follow from [AV02, Thm. 1.4.1].

We denote by $\mathcal{K}_g^{(1)}(\mathcal{R}/\mathcal{Y}) \subset \mathcal{K}_g(\mathcal{R}/\mathcal{Y})$ the open and closed substack of points $f : \mathcal{S} \to \mathcal{R}_T$ such that for all $t \in T$, the induced morphism $\overline{f}_t : S_t \to C_t$ of prestable curves of genus g is an isomorphism over a dense open subset of C_t [HLH25, Lem. 3.2]. Note that this actually forces $\overline{f} : S \to C$ to be an isomorphism. Hence the assignment Ψ above induces a proper and quasifinite morphism

$$\mathcal{K}_g^{(1)}(\mathcal{R}/\mathcal{Y}) \to \mathcal{Y}, \quad [f : \mathcal{S} \to \mathcal{R}_T] \mapsto [T \to \mathcal{Y}].$$

By construction the stack $\mathcal{K}_g(\mathcal{R}/\mathcal{Y})$ sends a scheme T to the groupoid of pairs $(\mathcal{S}, \mathcal{M})$ where

- $\mathcal{S} \to T$ is T-family of genus g twisted curves over with coarse space C_T,
- $\mathcal{M}$ is a line bundle on $\mathcal{S}$ such that the induced morphism $\mathcal{S} \to B\mathbb{G}_m$ is representable.
- $\mathcal{M}^{\otimes N}$ descends to the line bundle M_T on the coarse space C_T.

We may view the pair $(\overline{\mathcal{C}}^\circ, \mathcal{M}^\circ)$ as a morphism $\mathcal{Y}^\circ \to \mathcal{K}_g^{(1)}(\mathcal{R}/\mathcal{Y})$ fitting into the following commutative diagram of solid arrows

$$\begin{array}{ccc} \mathcal{Y}^\circ & \longrightarrow & \mathcal{K}_g^{(1)}(\mathcal{R}/\mathcal{Y}) \\ \downarrow & \nearrow & \downarrow \\ \mathcal{Y} & \longrightarrow & \mathcal{Y}. \end{array} \tag{3}$$

Since the morphism $\mathcal{K}_g^{(1)}(\mathcal{R}/\mathcal{Y}) \to \mathcal{Y}$ is proper and relatively DM, it follows that there exists a dotted arrow as in (3) making the diagram commute [AHLH23, Prop. 3.23, Prop. 3.46]. Therefore, we get an extension $(\overline{\mathcal{C}}, \mathcal{M})$ of the pair $(\overline{\mathcal{C}}^\circ, \mathcal{M}^\circ)$. By construction, the relative coarse space morphism $\overline{\mathcal{C}} \to C \to \mathcal{Y}$ is an isomorphism over the markings $\sigma_\bullet$. By taking appropriate root stacks of $\overline{\mathcal{C}}$ along the markings $\sigma_\bullet$, we obtain a family $(\mathcal{C}, \Sigma_\bullet)$ of marked twisted semistable curves over $\mathcal{Y}$ extending $(\mathcal{C}^\circ, \Sigma_\bullet)$. Setting $\mathcal{L} := \mathcal{M}|_{\mathcal{C}}(-\sum_i a_i \Sigma_i)$, we obtain the desired family $(\mathcal{C}, \Sigma_\bullet, \mathcal{L}) : \mathcal{Y} \to \mathcal{X}_{g,n}$ extending $(\mathcal{C}^\circ, \Sigma_\bullet^\circ, \mathcal{L}^\circ)$. □

Lemma 2.2.6. *Let $\mathcal{X}$ be an algebraic stack with affine diagonal, and consider a morphism $\mathcal{Y}^\circ \to \mathcal{X}$ for a test stack $\mathcal{Y}^\circ \subset \mathcal{Y}$ as in Definition 2.2.1 (1) or (2). Let $Y \to \mathcal{Y}$ (resp. $Y^\circ \to \mathcal{Y}^\circ$) be the quotient by $\mathbb{G}_m$ as in Notation 2.2.3. If the morphism $Y^\circ \to \mathcal{Y}^\circ \to \mathcal{X}$ extends to $Y \to \mathcal{X}$, then $\mathcal{Y}^\circ \to \mathcal{X}$ extends to $\mathcal{Y} \to \mathcal{X}$.*

Proof. Consider the presentation of $\mathcal{Y}$ given by

$$\mathbb{G}_m \times \mathbb{G}_m \times Y \rightrightarrows \mathbb{G}_m \times Y \rightrightarrows Y,$$

and the corresponding presentation for $\mathcal{Y}^\circ$. By assumption, we have an object $x \in \mathcal{X}(Y)$ which extends the object $x^\circ \in \mathcal{X}(Y^\circ)$. From the presentation of $\mathcal{Y}^\circ$, there is an isomorphism ξ° between the images of x° via the two maps $\mathcal{X}(Y^\circ) \rightrightarrows \mathcal{X}(Y^\circ \times \mathbb{G}_m)$, which we denote by x_1° and x_2°. These two images can also be obtained as the image of x via the following composition

$$\mathcal{X}(Y) \rightrightarrows \mathcal{X}(Y \times \mathbb{G}_m) \to \mathcal{X}(Y^\circ \times \mathbb{G}_m).$$

Denote by x_i the image of x via $\mathcal{X}(Y) \rightrightarrows \mathcal{X}(Y \times \mathbb{G}_m)$, which restricts to x_i°. From [DLI22, Lemma 4.3], the isomorphism $\xi^\circ \colon x_1^\circ \to x_2^\circ$ extends uniquely to an isomorphism $\xi \colon x_1 \to x_2$, as $\mathcal{X}$ has affine diagonal. The cocycle condition is again satisfied, from the uniqueness of the filling. Then there is a commutative diagram inducing $\mathcal{Y} \to \mathcal{X}$ as follows

$$\mathcal{X}(Y) \rightrightarrows \mathcal{X}(\mathbb{G}_m \times Y) \;\substack{\to\\ \to\\ \to}\; \mathcal{X}(\mathbb{G}_m \times \mathbb{G}_m \times Y). \qquad \square$$

Main result of the section

Theorem 2.2.7. *The stacks $\mathcal{X}_{g,n}$ and $\mathcal{A}nc_{g,n}$ are Θ-complete and S-complete. The same holds for the open substacks $\mathcal{X}_{g,n}^{\mathrm{rep}}$ and $\mathcal{A}nc_{g,n}^{\mathrm{rep}}$.*

Proof. By Lemma 2.2.4, it suffices to show that both $\mathcal{X}_{g,n}$ and $\mathcal{X}_{g,n}^{\mathrm{rep}}$ are S-complete and Θ-complete. In view of Lemma 2.2.5, it is enough to show that $\mathcal{X}_{g,n}^{\mathrm{rep}}$ is S-complete and Θ-reductive. This is the content of Theorem 2.5.3 below. $\square$

The rest of this section is dedicated to proving the Θ-completeness and S-completeness of $\mathcal{X}_{g,n}^{\mathrm{rep}}$, which was used in the proof of Theorem 2.2.7. In particular, for the rest of this section we work with non-twisted families of curves, which are fiberwise schemes.

2.3 Families of curves with regular total space

Lemma 2.3.1. *Let R be a DVR with fraction field K, and let T be either $\mathrm{Spec}(R)$ or $\mathbb{A}^1_K$. Consider a family of nodal curves $C \to T$ with smooth generic fiber. Then there is a unique family of nodal curves $C' \to T$ with a birational morphism $\varphi' \colon C' \to C$ over T, such that:*

(1) C' is regular, and

(2) $\omega_{C'/T}$ is nef relative to C.

Moreover, the morphism φ' is an isomorphism away from the singular locus of C, and $C' \to C$ is minimal in the following sense: given any family of nodal curves $C'' \to T$ with a birational morphism $\varphi'' \colon C'' \to C$ over T and with C'' regular, there is a morphism $b \colon C'' \to C'$ such that $\varphi'' = \varphi' \circ b$.

Remark 2.3.2. The morphism $C' \to C$ of Lemma 2.3.1 is a *minimal resolution*; see [Das15, Thm. 2.8].

Remark 2.3.3. Let R be a DVR, let T be either $\mathrm{Spec}(R)$ or $\mathbb{A}^1_K$, and consider a family of marked nodal curves $(C, \sigma_\bullet) \to T$ with smooth generic fiber, as in Lemma 2.3.1. Since the family of curves $C' \to C$ of Lemma 2.3.1 is an isomorphism on the smooth locus of $C \to T$, the sections $\sigma_\bullet$ lift to sections $\sigma'_\bullet$ of $C' \to T$.

Proof of Lemma 2.3.1. We provide a proof when $T = \operatorname{Spec}(R)$; the argument for $T = \mathbb{A}^1_K$ is similar. Let $p \in C$ be a singular point of the total space of C. Since $C \to \operatorname{Spec}(R)$ is a family of nodal curves, from [Sta25, Tag 0CDD] there is an isomorphism

$$\mathcal{O}^{sh}_{C,p} \cong (R[s,t]/(st - \pi^m))^{sh}$$

where $m \geq 1$ and π is the uniformizer of R. One can check by an explicit computation that:

1. to resolve the singularity of $\operatorname{Spec}(R[s,t]/(st-\pi^m))$ when $m > 1$, one can consecutively perform a blow-up of the ideal sheaf of the singular point, with the reduced structure; and
2. the fiber over $\pi = 0$ has nodal singularities.

The property of being regular and nodal over $\operatorname{Spec}(R)$ are local in the étale topology [Sta25, Tag 0C5D] and blow-ups commute with flat (hence étale) base change [Sta25, Tag 085S], so the blow-ups of C at its singular points, with the reduced structure, will lead to a family of nodal curves $C' \to \operatorname{Spec}(R)$ with C' regular.

To check that $\omega_{C'/\operatorname{Spec}(R)}$ is fiberwise trivial over C, it suffices to show that its restriction to every exceptional divisor has degree 0. This can again be checked for $\operatorname{Spec}(R[s,t]/(st-\pi^m))$, where it is an explicit computation.

We argue now that $C' \to C$ is the minimal resolution of [Das15, Thm. 2.8]; which will conclude the argument. Let $C^m \to C$ be the minimal resolution of [Das15, Thm. 2.8] and let $\xi \colon C' \to C^m$ be the map of [Das15, Thm. 2.8]. From [Liu02, Thm. 9.2.2], the map ξ is a composition of ℓ-many blow-ups along closed points. Let's assume by contradiction that $\ell > 0$ and let E be the last exceptional divisor extracted by these blowups. Then $E^2 = -1$ and $\omega_{C'/\operatorname{Spec}(R)}(E)|_E \cong \mathcal{O}^1_{\mathbb{P}}(-2)$ so

$$\omega_{C'/\operatorname{Spec}(R)}.E < 0.$$

But E is also contracted by $C' \to C$, which contradicts that $\omega_{C/\operatorname{Spec}(R)}$ is nef over C'. □

Lemma 2.3.4. *Let R be a DVR with algebraically closed residue field, and let $\mathcal{Y}$ be either $\overline{ST}_R$ or Θ_R. Let $\widehat{C}$ be a family of nodal curves over $\operatorname{Spec}(R)$ with smooth generic fiber. Then there exists a family of nodal curves $C \to \mathcal{Y}$ such that:*

(a) C is regular and $\omega_{C/\operatorname{Spec}(R)}$ is fiberwise trivial over $\widehat{C} \times_{\operatorname{Spec}(R)} \mathcal{Y}$.

(b) There is a morphism $C \to \widehat{C} \times_{\operatorname{Spec}(R)} \mathcal{Y}$ which contracts only rational bridges.

In particular, if $\widehat{C}$ is stable, then $C \to \mathcal{Y}$ is semistable.

Proof. Consider the family of curves $\widetilde{C} \to \widehat{C}$ of Lemma 2.3.1, and let C' be the fiber product

$$\begin{array}{ccc} C' & \longrightarrow & \widetilde{C} \\ \downarrow & & \downarrow \\ \mathcal{Y} & \longrightarrow & \operatorname{Spec}(R). \end{array}$$

If $\mathcal{Y} = \Theta_R$, the morphism $\mathcal{Y} \to \operatorname{Spec}(R)$ is smooth, so its pullback $C' \to \widetilde{C}$ is also smooth. Since $\widetilde{C}$ is regular, we have that C' is regular.

If instead $\mathcal{Y} = \overline{ST}_R$, then C' is not regular. To conclude the proof, we claim that there is a family of nodal curves $C \to \overline{ST}_R$ with total regular space and equipped with a proper birational morphism $C \to C'$ that is an isomorphism away from

the singular points of $C'_{\mathfrak{o}}$. Let $Z \subset C'$ denote the union of the preimages of the nodes in the special fiber of $\widetilde{C}$. Note that the sought-after modification $C \to C'$ will be an isomorphism over the complement of Z. By Proposition A.0.1, in order to construct such modification $C \to C'$, we may replace C' with the base-change $C' \times_{\widetilde{C}} \operatorname{Spec}(\widehat{\mathcal{O}}_{\widetilde{C},p})$ over the completion of $\widetilde{C}$ at each of the nodes $p \in \widetilde{C}$ in the special fiber. Since the residue field of $\operatorname{Spec}(R)$ is algebraically closed, we have an isomorphism $\widehat{\mathcal{O}}_{\widetilde{C},p} \cong \widehat{R}[\![x,y]\!]/(xy-\pi)$ [Sta25, Tag 0CBX], and hence an isomorphism

$$C' \times_{\widetilde{C}} \operatorname{Spec}(\widehat{\mathcal{O}}_{\widetilde{C},p}) \cong \operatorname{Spec}(\widehat{R}[\![x,y]\!][s,t]/(xy-st))/\mathbb{G}_m.$$

where $\mathbb{G}_m$ acts trivially on $\widehat{R}[\![x,y]\!]$ and with the usual weights on the coordinates s and t. To conclude, we set $C := B/\mathbb{G}_m$, where B is the blowup of $\operatorname{Spec}(\widehat{R}[\![x,y]\!][s,t]/(xy-st))$ at the $\mathbb{G}_m$-equivariant closed subscheme cut out by the ideal (x,s). The fact that B is regular, $\mathcal{Y}$-flat and has nodal $\mathcal{Y}$-fibers follows from direct computation. □

For the following result we use the terminology in Notation 2.2.3.

Corollary 2.3.5. *Let R be DVR with algebraically closed residue field, and let $\mathcal{Y}$ be either $\overline{ST}_R$ or Θ_R. Let $(C^\circ, \sigma_\bullet^\circ)$ be a family of n-marked semistable curves over $\mathcal{Y}^\circ$, whose stabilization is a family of nodal marked curves. Suppose that the total space C° is regular. Then, there exists a family $(C, \sigma_\bullet)$ of n-marked semistable curves over $\mathcal{Y}$ extending $(C^\circ, \sigma_\bullet^\circ)$ such that C is regular.*

Note that the assumption that the stabilization is a family of nodal pointed curves is not satisfied only when C° has genus 0 and $n < 3$, or when C° has genus 1 and $n = 0$.

Proof. The stabilization of $(C^\circ, \sigma_\bullet^\circ)$ corresponds to a family $\mathcal{Y}^\circ \to \overline{\mathcal{M}}_{g,n}$. Since $\overline{\mathcal{M}}_{g,n}$ is separated and has finite inertia, this extends to a family of stable curves $\mathcal{Y} \to \overline{\mathcal{M}}_{g,n}$ [AHLH23, Prop. 3.23, Prop. 3.46]. Furthermore, the morphism $\mathcal{Y} \to \overline{\mathcal{M}}_{g,n}$ factors through $\mathcal{Y} \to \operatorname{Spec}(R)$ by [AHLH23, Lem. 3.6, Lem. 3.45]. In other words, there is a family $(C^{\mathrm{st}}, \sigma_\bullet^{\mathrm{st}})$ over $\operatorname{Spec}(R)$ whose base-change to $\mathcal{Y}^\circ$ recovers the stabilization of $(C^\circ, \sigma_\bullet^\circ)$. By Lemma 2.3.4, there is a family $(\widetilde{C}, \widetilde{\sigma}_\bullet)$ of marked nodal curves with $\widetilde{C} \to \mathcal{Y}$ regular and such that the stabilization of $(\widetilde{C}, \widetilde{\sigma}_\bullet)$ is isomorphic to $(C^{\mathrm{st}}, \sigma_\bullet^{\mathrm{st}}) \times_{\operatorname{Spec}(R)} \mathcal{Y}$.

Consider the pulled-back families

$$(C_1, \sigma_{\bullet,1}) := (\widetilde{C}, \widetilde{\sigma}_\bullet) \times_{\mathcal{Y}} Y,\ (C_1^\circ, \sigma_{\bullet,1}^\circ) := (\widetilde{C}, \widetilde{\sigma}_\bullet) \times_{\mathcal{Y}} Y^\circ \text{ and } (C_2^\circ, \sigma_{\bullet,2}^\circ) := (C, \sigma_\bullet) \times_{\mathcal{Y}} Y^\circ$$

where Y is as in Notation 2.2.3. In particular, Y° is the union of two schemes which are either two copies of $\operatorname{Spec}(R)$, or a a copy of $\mathbb{G}_m \times \operatorname{Spec}(R))$ and $\mathbb{A}^1_K$, along common open subsets. From the minimality in Lemma 2.3.1, and since the family $\widetilde{C}$ satisfies that $\omega_{\widetilde{C}/\mathcal{Y}}$ is fiberwise trivial over $C^{\mathrm{st}} \times_{\operatorname{Spec}(R)} \mathcal{Y}$ there is a map $C_2^\circ \longrightarrow C_1^\circ$.

From [Liu02, Thm. 9.2.2] this map is a composition of blow-ups. Since by assumption $(C_2^\circ, \sigma_{\bullet,2}^\circ) \to Y^\circ$ is semistable with smooth generic fiber, the divisor $\omega_{C_2^\circ}(\sigma_{\bullet,2}^\circ)$ is nef relative to Y°. Therefore these blow-ups are as follows. If $Y = \operatorname{Spec}(R[s,t]/(st-\pi))$, they are supported at the intersection of the closed points of Y° and some of the sections $\sigma_{\bullet,2}^\circ$. If instead $Y = \operatorname{Spec}(R[t])$, these blow-ups are supported at the intersection of some of the sections $\sigma_{\bullet,2}^\circ$ and the locus in Y° given by $\pi \cdot t = 0$. Let

$$B_\ell^\circ = C_2^\circ \to B_{\ell-1}^\circ \to \ldots \to B_1^\circ = C_1^\circ$$

be these blowups, and let $\Delta_1^\circ, \dots, \Delta_\ell^\circ$ be the loci we are blowing up, with $\Delta_i^\circ \subseteq B_i^\circ$. We now construct extensions $B_i \to Y$ of the family B_i° by setting $B_1 = C_1$ and iteratively blowing up loci inside the closure of Δ_{i-1}° in B_{i-1}. One can proceed by induction to argue that Δ_i, the closure of Δ_i° in B_i, is regular. The sections $\sigma_{\bullet,2}^\circ$ extend to sections of $B_\ell \to Y$ which we denote by $\sigma_{\bullet,B}$. The pair $(B_\ell, \sigma_{\bullet,B}) \to S$ is a Y-family of pointed curves extending $(C_2^\circ, \sigma_{\bullet,2}^\circ)$.

Finally observe that these blow-ups are all blow-ups of $\mathbb{G}_m$-equivariant loci, so the action of $\mathbb{G}_m$ on $(B_\ell, \sigma_{\bullet,B})|_{Y^\circ}$ extends to an action on $(B_\ell, \sigma_{\bullet,B})$, giving the desired extension $(C, \sigma_\bullet)$ over $\mathcal{Y}$. □

Lemma 2.3.6. *Let R be DVR, and let $\mathcal{Y}$ be either $\overline{ST}_R$ or Θ_R. Let $(C^\circ, \sigma_1^\circ, \sigma_2^\circ)$ be a family of 2-marked genus 0 semistable curves over $\mathcal{Y}^\circ$. Suppose that the total space C° is regular. Then, there exists a 2-marked family (C, σ_1, σ_2) of semistable curves over $\mathcal{Y}$ extending $(C^\circ, \sigma_1^\circ, \sigma_2^\circ)$ such that C is regular.*

Proof. We use the notation of Notation 2.2.3. We write u, v for the pair of parameters cutting out $\mathfrak{o} \in \mathcal{Y}$. The stack $\mathcal{Y}^\circ$ is the union of the two opens

$$\mathcal{Y}_u := \{v \neq 0\}/\mathbb{G}_m \qquad \text{and} \qquad \mathcal{Y}_v := \{u \neq 0\}/\mathbb{G}_m$$

which are glued along $\mathrm{Spec}(K)$. If $\mathcal{Y} = \overline{ST}_R$, then $\mathcal{Y}_u \cong \mathcal{Y}_v \cong \mathrm{Spec}(R)$. If $\mathcal{Y} = \Theta_R$, then $\mathcal{Y}_u \cong \mathrm{Spec}(R)$ and $\mathcal{Y}_v \cong \Theta_K$. In the latter case, we work $\mathbb{G}_m$-equivariantly on $\mathrm{Spec}(K[t])$, whose point $t = 0$ lies over the closed point of Θ_K. Over each of $\mathcal{Y}_u, \mathcal{Y}_v$, the special fibre F of C° is a chain $R_1 \cup \ldots \cup R_m$ of rational curves, with σ_1° on R_1 and σ_2° on R_m.

Choose one of the two branches, and suppose $m \geq 2$. The rational curve R_m meets the rest of fibre F in a single node of smoothing order one, so

$$R_m.(F - R_m) = 1 \text{ and } F.R_m = 0, \text{ giving } R_m^2 = -1.$$

We apply Castelnuovo's criterion iteratively, contracting $R_m, R_{m-1}, \dots, R_2$, and $\sigma_2'^\circ$ is sent to the image of the node $R_m \cap R_{m-1}$ each time. Doing the same on the other branch, together with the fact that they agree over $\mathrm{Spec}(K)$ where no contractions are performed, gives a $\mathbb{P}^1$-bundle over $\mathcal{Y}^\circ$ equipped with two markings.

This extends to a $\mathbb{P}^1$-bundle with two markings over $\mathcal{Y}$ since the stack of projective bundles with two markings is isomorphic to $B\mathbb{G}_m$, which is S-complete and Θ-complete. We denote the corresponding $\mathbb{G}_m$-equivariant family over Y by $(C^{\mathrm{st}}, \sigma_1^{\mathrm{st}}, \sigma_2^{\mathrm{st}})$, and take D^{st} to be the divisor associated to the marked point σ_2^{st}. This plays the role of the stabilization.

We now realize the contraction $C^\circ \to C^{\mathrm{st}}|_{\mathcal{Y}^\circ}$ as a sequence of blowups, similar to blow-ups of Corollary 2.3.5. On each branch of $\mathcal{Y}^\circ$, this contraction is a proper birational morphism of regular surfaces, contracting a chain of rational curves onto the point of D^{st} over the closed point of the branch. Applying [Sta25, Tag 0C5Q] shows that $C^\circ \to C^{\mathrm{st}}|_{\mathcal{Y}^\circ}$ is an iterated sequence of blow-ups at closed points lying over $D^{\mathrm{st}}|_{u=0}$ and $D^{\mathrm{st}}|_{v=0}$. We define $C \to C^{\mathrm{st}}$ to be the iterated sequence of blowups over all of $\mathcal{Y}$, where we take the centers to be the corresponding flat closures of the centers over $\mathcal{Y}^\circ$. Note that C is a $\mathbb{G}_m$-equivariant family of prestable curves over Y with regular total space, since the centers of the blowups always lie in smooth loci over Y. The markings $\sigma_1^\circ, \sigma_2^\circ$ extend uniquely as sections σ_1, σ_2 over $\mathcal{Y}$ and by construction, we have that the restriction of (C, σ_1, σ_2) to $\mathcal{Y}^\circ$ recovers $(C^\circ, \sigma_1^\circ, \sigma_2^\circ)$. □

2.4 Birational modifications of families of curves

We gather here some further preliminary results needed for our proof Θ-completeness and S-completeness.

2.4.1 Contracting rational curves

Lemma 2.4.1. *Let $(C, \sigma_\bullet) \to T$ be a family of n-marked semistable curves of genus $g \geq 2$ over an algebraic stack T, and let $f : (C, \sigma_\bullet) \to (C^{\mathrm{st}}, \sigma_\bullet^{\mathrm{st}})$ denote its stabilization. Suppose L is a line bundle on C that is fiberwise nef with respect to f. Then the following hold:*

(1) L is relatively semi-ample for f.

(2) The scheme $\mathrm{Proj}_{C^{\mathrm{st}}}(\oplus_{m\geq 0} f_*(L^{\otimes m}))$ equipped with the images of the markings $\sigma_\bullet$ is a family of n-marked semistable curves over T, and its formation commutes with T-base change.

Proof. We work smooth-locally on T and assume that T is an affine scheme. Furthermore, by [Sta25, Tag 0DQ0], after passing to an étale cover of T, we may assume that $C \to T$ admits a relative polarization, and so in particular C is a scheme. By standard approximation results, we may furthermore assume that T is Noetherian. It suffices to show that the following hold:

(a) For all $m \geq 0$, the pushforward $f_*(L^{\otimes m})$ is T-flat and its formation commutes with T-base-change.

(b) For $m \gg 0$, the counit $f^* f_*(L^{\otimes m}) \to L^{\otimes m}$ is surjective.

(c) For all $t \in T$, the t-fiber of $\mathrm{Proj}_{C^{\mathrm{st}}}(\oplus_{m\geq 0} f_*(L^{\otimes m}))$ is a prestable curve.

By relative cohomology and base-change as in [FHI26, Prop. 3], to check (a) it suffices to show that for all $t \in T$ the base-change $f_t : C_t \to C_t^{\mathrm{st}}$ satisfies $R^1(f_t)_*(L_t^{\otimes m}) = 0$ for all $m \geq 0$. Once we check this, then (b) and (c) can be checked fiberwise over T. Therefore we may assume without loss of generality that $T = \mathrm{Spec}(k)$. Now to check (b), (c) and the vanishing of $R^1 f_*(L^{\otimes m})$, we may work étale locally around a neighborhood of each node of C^{st}. Then the result follows from a local computation where the fibres are chains of rational curves adjoined with affine opens. In this case the resulting semistable curve $\mathrm{Proj}_{C^{\mathrm{st}}}(\oplus_{m\geq 0} f_*(L^{\otimes m}))$ is obtained by contracting the rational curves in the chain where L restricts trivially. □

2.4.2 Flipping rational curves

Example 2.4.2. We recall how to construct the Atiyah flop. Consider the cone of a quadric $\mathrm{Spec}(A)$, where $A = K[x, y, s, t]/(xy - st)$ over a field K. Let B_1 be the blow-up of $V(x, s)$, and B_2 the blow-up of $V(x, t)$. An explicit computation shows that the corresponding blow-up morphisms $B_i \to \mathrm{Spec}(A)$ are small and contract a single curve isomorphic to $\mathbb{P}^1$, and that B_i are smooth. We will refer to the birational transformation $B_1 \dashrightarrow B_2$ as the *Atiyah flop*. We denote by $\widehat{B}_1 \to \mathrm{Spec}(\widehat{A}) \leftarrow \widehat{B}_2$ the base-change to the completion $\widehat{A}$ of A at the origin $V(x, y, s, t)$.

Observe also that, if we denote by R_i the curve isomorphic to $\mathbb{P}^1$ in B_i contracted by $B_i \to \mathrm{Spec}(A)$, then the blow-up of B_i along R_i is isomorphic to the blow-up of $\mathrm{Spec}(A)$ at $V(x, y, s, t)$.

Lemma 2.4.3. *Let $X \to Y$ be a family of semistable curves over a regular integral Noetherian scheme Y of dimension 2. Suppose that there is a closed point $\mathfrak{o} \in Y$ such that the fiber $X_{\mathfrak{o}}$ contains a rational bridge $R \cong \mathbb{P}^1_{\kappa(\mathfrak{o})} \hookrightarrow X$, such that X is regular in a neighborhood of R, and the normal bundle satisfies $\mathcal{N}_{R/X} \cong \mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$. Then there exist:*

- *a family $X' \to Y$ of semistable curves,*
- *a rational curve $R' \cong \mathbb{P}^1 \hookrightarrow X'_{\mathfrak{o}}$,*

- *Y-morphisms $\pi : X \to X^0$ and $\pi' : X' \to X^0$ to a proper normal Y-algebraic space X^0, and*
- *a $\kappa(\mathfrak{o})$-point $p \in X^0_{\mathfrak{o}}$,*

satisfying the following:

(1) We have $\pi(R) = \pi'(R') = \{p\}$, and the morphisms π and π' are isomorphisms over $X^0 \setminus \{p\}$.

(2) There is an identification $\widehat{\mathcal{O}}_{X^0,p} \cong \widehat{A}$ with the completion of the quadric cone at the origin over the field $K = \kappa(\mathfrak{o})$ as in Example 2.4.2, and a commutative diagram where the two squares are Cartesian

$$\begin{array}{ccccc} \widehat{B}_1 & \longrightarrow & \operatorname{Spec}(\widehat{A}) & \longleftarrow & \widehat{B}_2 \\ \downarrow & & \downarrow & & \downarrow \\ X & \longrightarrow & X^0 & \longleftarrow & X'. \end{array}$$

(3) Let $C \hookrightarrow X_{\mathfrak{o}} \subset X$ be another curve that meets R in a single node of $X_{\mathfrak{o}}$, and let $C' \hookrightarrow X'$ denote the strict transform. For any invertible sheaf L on X, if L' denotes the corresponding invertible sheaf on X', then

$$\deg(L'|_{R'}) = -\deg(L|_R) \qquad \text{and} \qquad \deg(L'|_{C'}) = \deg(L|_C) + \deg(L|_R). \quad (4)$$

Proof. We first prove (1) and (2). The statement is local around $\mathfrak{o}$, so up to replacing Y with a Zariski neighbourhood of it we can add an effective Cartier divisor D on X, so that $\omega_{X/Y}(D)$ is nef and the only irreducible of $X_{\mathfrak{o}}$ over which $\omega_{X/Y}(D)$ is zero is R. Observe that, from the explicit description of the normal bundle of R, the curve R is rigid. Lemma 2.4.1 gives a family of nodal curves which, since R is rigid, contracts R to a point and is an isomorphism away from R; let X^0 be such a family and let p be the point to which R is contracted.

As $X^0 \to Y$ is a family of nodal curves, from [Sta25, Tag 0CBX] we have

$$\widehat{\mathcal{O}}_{X^0,p} \cong \widehat{\mathcal{O}}_{Y,\mathfrak{o}}[\![s,t]\!]/(st - f)$$

for $f \in \widehat{\mathcal{O}}_{Y,\mathfrak{o}}$. We now argue that f is the product of two generators of the maximal ideal in $\widehat{\mathcal{O}}_{Y,\mathfrak{o}}$. For each node n_1, n_2 of R the result [Sta25, Tag 0CBX] gives:

$$\widehat{\mathcal{O}}_{X,n_1} \cong \widehat{\mathcal{O}}_{Y,\mathfrak{o}}[\![s_1,t_1]\!]/(s_1t_1 - f_1) \text{ and } \widehat{\mathcal{O}}_{X,n_2} \cong \widehat{\mathcal{O}}_{Y,\mathfrak{o}}[\![s_2,t_2]\!]/(s_2t_2 - f_2). \quad (5)$$

Since X is regular, the elements f_i are non-zero in $\mathfrak{m}_{\mathfrak{o}}/\mathfrak{m}^2_{\mathfrak{o}}$. Their vanishing locus in $\operatorname{Spec}(\widehat{\mathcal{O}}_{Y,\mathfrak{o}})$ is the image of the closure of the connected components of the singular locus of the morphism

$$\mathfrak{X} := X \times_Y \operatorname{Spec}(\widehat{\mathcal{O}}_{Y,\mathfrak{o}}) \to \operatorname{Spec}(\widehat{\mathcal{O}}_{Y,\mathfrak{o}}) =: \mathfrak{Y}$$

containing n_1 and n_2. We now argue that f_1 and f_2 generate the maximal ideal $\mathfrak{m}_{\mathfrak{o}}$.

Assume by contradiction that they do not, consider $g \in \mathfrak{m}_{\mathfrak{o}}$ such that f_1, g generate $\mathfrak{m}_{\mathfrak{o}}/\mathfrak{m}^2_{\mathfrak{o}}$. Consider the following commutative diagram

$$\begin{array}{ccccccccc} 0 & \longrightarrow & I_R & \longrightarrow & \mathcal{O}_X & \longrightarrow & \mathcal{O}_R & \longrightarrow & 0 \\ & & \uparrow{\scriptstyle \alpha} & & \uparrow & & \uparrow & & \\ 0 & \longrightarrow & \mathfrak{m}_{\mathfrak{o}} & \longrightarrow & \mathcal{O}_{Y,\mathfrak{o}} & \longrightarrow & k & \longrightarrow & 0. \end{array}$$

The image of f_1 via α gives a section of I_R/I_R^2, which from (5) is non-zero, and gives an exact sequence

$$0 \to \mathcal{O}_R \to I_R/I_R^2 \to K \to 0$$

where K is the cokernel. An explicit computation using (5) and the fact that f_1 and f_2 have proportional image in I_R/I_R^2 shows that K splits as $\mathcal{O}_R \oplus k_{n_1} \oplus k_{n_2}$, where k_{n_i} is the skyscraper sheaf at n_i, and the image of g in I_R/I_R^2 is a global section for $\mathcal{O}_R$. In particular, there is a surjection $I_R/I_R^2 \cong \mathcal{O}_{\mathbb{P}^1}(1) \oplus \mathcal{O}_{\mathbb{P}^1}(1) \to \mathcal{O}_{\mathbb{P}^1}$, which is a contradiction, as such a surjection does not exist on $\mathbb{P}^1$.

Since the curve we are contracting is rigid, the morphism $X \to X^0$ is an isomorphism away from $\mathfrak{o}$; we now argue that, up to replacing f with $u \cdot f$ for a unit u, we have $f = f_1 \cdot f_2$. Since $\mathfrak{Y}$ is S_2, it suffices to show the desired equality on $\mathfrak{Y} \smallsetminus \mathfrak{o}$. The equality follows since the set of points in $\mathfrak{Y} \smallsetminus \mathfrak{o}$ satisfying $\{f = 0\}$ agrees with the image of the connected components of the nodal locus of

$$\mathfrak{X} \smallsetminus \mathfrak{X}_\mathfrak{o} \to \mathfrak{Y} \smallsetminus \mathfrak{o}$$

whose closure in $\mathfrak{X}$ contains n_1 or n_2. The latter is $\{f_1 \cdot f_2 = 0\}$, so since $\mathfrak{Y}$ is locally factorial, we have $f = f_1^a \cdot f_2^b \cdot u$ for u a unit. Since the contraction $X \to X^0$ is an isomorphism away from R, the scheme X^0 is regular in a neighbourhood of p away from p, so again a local computation shows $a = b = 1$. This shows that

$$\widehat{\mathcal{O}}_{X^0,p} \cong \widehat{\mathcal{O}}_{Y,\mathfrak{o}}[\![s,t]\!]/(st - f_1 f_2) \cong \kappa(\mathfrak{o})[\![x,y,z,t]\!]/(xy - zt), \tag{6}$$

where for the last isomorphism we used [Sta25, Tag 0C0S].

Observe now that

$$\mathrm{Pic}(\mathfrak{X}^0 \smallsetminus \{p\}) = \mathbb{Z}$$

where we denote by $\mathfrak{X}^0 := \mathrm{Spec}(\widehat{\mathcal{O}_{X^0,p}})$, so if $\mathcal{L}$ is a generator of such a group, since X is regular, it extends to a line bundle $\mathcal{M}$ on $\widehat{\mathfrak{X}} := X \times_{X^0} \mathfrak{X}^0$. Since $\widehat{\mathfrak{X}} \to \mathfrak{X}^0$ is projective, we have that

$$\widehat{\mathfrak{X}} = \mathrm{Proj}\left(\bigoplus_{n \geq 0} H^0(\widehat{\mathfrak{X}}, \mathcal{M}^n)\right) \text{ or } \widehat{\mathfrak{X}} = \mathrm{Proj}\left(\bigoplus_{n \leq 0} H^0(\widehat{\mathfrak{X}}, \mathcal{M}^n)\right).$$

In particular, after identifying $\widehat{\mathcal{O}}_{X^0,p}$ with $\kappa(\mathfrak{o})[\![x,y,z,t]\!]/(xy - zt)$ via Equation (6), we can identify the two Proj above with $\widehat{B}_1$ and $\widehat{B}_2$, and up to replacing B_1 and B_2, we can assume $\widehat{\mathfrak{X}} \cong \widehat{B}_1$ over $\mathrm{Spec}(\widehat{\mathcal{O}}_{X^0,p})$.

Applying [Sta25, Tag 0GDM] to the map $B_2 \to \mathrm{Spec}(\kappa(\mathfrak{o})[x,y,s,t]/xy - st)$, and using again Equation (6), gives a formal modification of X^0, which from [Sta25, Tag 0GDU] gives the birational map $X' \to X^0$. These maps agree over $\mathfrak{X}^0$. In particular, X' is regular in a neighbourhood of R'. Therefore, in a neighbourhood of R', the morphism $X' \to Y$ is a morphism of regular schemes with equidimensional fibers of dimension 1, and hence it is flat by miracle flatness. By the explicit description of the flop, the map $X' \to Y$ is nodal. This concludes the argument for (1) and (2).

Observe also that $\mathrm{Bl}_R(X) \cong \mathrm{Bl}_{R'}(X')$, where $\mathrm{Bl}_R(X)$ (resp. $\mathrm{Bl}_{R'}(X')$) is the blow-up of X at R (resp. X' at R'). Indeed, it suffices to show that they are isomorphic over $\mathrm{Spec}(\widehat{\mathcal{O}}_{X^0,p})$, which follows from the previous identifications and Example 2.4.2.

For the degree identities, let $Z \cong \mathrm{Bl}_R(X) \cong \mathrm{Bl}_{R'}(X')$ with projections π and π' onto X and X' respectively. Here we identify the exceptional divisor $E \cong R \times R'$ so that $\pi|_E$ and $\pi'|_E$ are the two projections. For closed points $y \in R'$ and $x \in R$, the

ruling $R \times \{y\}$ maps isomorphically to R under π, and $\{x\} \times R'$ maps isomorphically to R' under π'. We know that

$$\pi^*(L) \cong (\pi')^*(L') \otimes \mathcal{O}_Z(aE) \tag{7}$$

for some $a \in \mathbb{Z}$. Taking the degree of both sides of (7) along the curve $R \times \{y\} \hookrightarrow E$ gives $\deg(L|_R) = a \deg(\mathcal{O}(E)|_{R\times\{y\}}) = -a$, because $\mathcal{O}(E)|_E \cong \mathcal{O}(-1,-1)$. On the other hand, restricting to $\{x\} \times R' \hookrightarrow E$ gives $0 = \deg(L'|_{R'}) - a$. This gives the first identity. For the second identity, we observe that because C meets R transversely at a node, the strict transform $\tilde{C} \hookrightarrow Z$ meets the exceptional divisor E transversely at a single point as well, and maps isomorphically to both C and C'. Therefore, $\deg(\mathcal{O}(E)|_{\tilde{C}}) = 1$, and restricting (7) to $\tilde{C}$ gives the second identity. □

Notation 2.4.4. We let Y denote either of the two test surfaces as in Notation 2.2.3 with closed point $\mathfrak{o}$ cut out by a pair of parameters, which we write uniformly as u, v:

$$(u,v) = (s,t) \text{ if } Y = \operatorname{Spec}(R[s,t]/(st-\pi)), \qquad (u,v) = (\pi,t) \text{ if } Y = \operatorname{Spec}(R[t]).$$

We set $Y_u := \{u = 0\}$ and $Y_v := \{v = 0\}$, so that $Y_u \cap Y_v = \mathfrak{o}$. Let $(C, \sigma_\bullet) \to Y$ be a $\mathbb{G}_m$-equivariant family of marked semistable curves such that the stabilization $(C^{\mathrm{st}}, \sigma_\bullet^{\mathrm{st}}) \to Y$ has smooth generic fiber. Since the family $(C, \sigma_\bullet) \to Y$ is $\mathbb{G}_m$-equivariant, the stabilization descends to a morphism $(C^{\mathrm{st}}, \sigma_\bullet^{\mathrm{st}}) : \mathcal{Y} \to \overline{\mathcal{M}}_{g,n}$ and since $\overline{\mathcal{M}}_{g,n}$ is separated and has finite inertia this morphism factors through $\mathcal{Y} \to \operatorname{Spec}(R)$ by [AHLH23, Lem. 3.6, Lem. 3.45]. Hence there is a family of marked stable curves $(C_R^{\mathrm{st}}, \sigma_{\bullet,R}^{\mathrm{st}}) \to \operatorname{Spec}(R)$ whose base-change to Y recovers $(C^{\mathrm{st}}, \sigma_\bullet^{\mathrm{st}})$.

Let z_R be either a node or a marking of the special fiber of C_R^{st}.

1. Suppose z_R is a node. From the description of the versal deformation space of nodal singularities as $xy = \alpha$ for some function α on the base Y [Sta25, Tag 0CBX], the locus over which z_R stays singular is $\{\alpha = \pi = 0\}$, and the preimage of $\pi = 0$ in Y is contained in $Y_u \cup Y_v$. If instead z_R is a marking, it extends to a section of $C^{\mathrm{st}} \to Y$ contained in the smooth locus.
2. Let ℓ and m be the numbers of irreducible components of the rational chains of C mapping to z_R over Y_u and Y_v respectively. We assume that the fibre $C_\mathfrak{o}$ has a rational chain over z_R with N irreducible components, where

$$N := \ell + m + 1 \text{ if } z_R \text{ is a node}, \qquad N := \ell + m \text{ if } z_R \text{ is a marking}.$$

   We label the special points of this rational chain by $x_0, \ldots, x_N$, where x_{i-1} and x_i are joined by the rational curve $R_i \cong \mathbb{P}^1$ for $1 \le i \le N$. If z_R is a node, every x_i is a node of $C_\mathfrak{o}$. If z_R is a marking, $x_0, x_1, \ldots, x_{\ell+m-1}$ are nodes, and x_N is the marking of $C_\mathfrak{o}$ lying over z_R.

Lemma 2.4.5. *In the setting of Notation 2.4.4, let $X \to Y$ be a family of nodal curves whose total space X is regular, and let $R_i \cong \mathbb{P}^1$ be a rational curve in the central fibre $X_\mathfrak{o}$, where x_i and x_{i-1} are both nodes of $X_\mathfrak{o}$, étale locally of the form $xy = f$ for some functions f on Y. Suppose that one of x_i, x_{i-1} is preserved over Y_u and the other is preserved over Y_v in the sense that $V(f) = Y_u$, respectively $V(f) = Y_v$. Then $\mathcal{N}_{R_i/X} \cong \mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$.*

Proof. After completing the DVR and replacing the ground field k with its residue field, we may assume for simplicity of notation that $\kappa(\mathfrak{o}) = k$. Suppose x_{i-1} is preserved over Y_u and x_i over Y_v. The two nodes are analytically locally of the form $\operatorname{Spec}(k[\![x,y,u,v]\!]/(xy - u^a))$ and $\operatorname{Spec}(k[\![x,y,u,v]\!]/(xy - v^b))$ as above. Since the

family of nodal curves has regular total space X, then u^a and v^b must vanish with multiplicity one along Y_u and Y_v respectively, forcing $a = b = 1$. Thus we have $\widehat{\mathcal{O}}_{X,x_{i-1}} \cong k[\![x, y, v]\!]$ with $u = xy$ and $\widehat{\mathcal{O}}_{X,x_i} \cong k[\![x, y, u]\!]$ with $v = xy$.

Let $\mathcal{I} \subset \mathcal{O}_X$ denote the ideal sheaf of R_i. We also use u and v to denote the corresponding sections of the rank two vector bundle $\mathcal{I}/\mathcal{I}^2 = \mathcal{I}|_{R_i}$ on R_i. Let us analyze these sections at the completions of closed points of R_i. For a smooth point $z \in R_i$ that is not x_i or x_{i-1}, we have $\widehat{\mathcal{O}}_{X,z} \cong k[\![u, v, w]\!]$ for a local fibre coordinate w. Near the point z, we have $\mathcal{I} = (u, v)$ and u, v are a basis of $\mathcal{I}/\mathcal{I}^2$. At the node x_{i-1}, we have $\mathcal{I} = (x, v)$ and x, v are a basis of $\mathcal{I}/\mathcal{I}^2$. Symmetrically at x_i, we have $\mathcal{I} = (u, y)$ and u, y are a basis of $\mathcal{I}/\mathcal{I}^2$.

Let $\mathcal{F}_u$ and $\mathcal{F}_v$ denote the saturations of the subsheaves of $\mathcal{I}/\mathcal{I}^2$ generated by u and v respectively. Locally, $\mathcal{F}_u$ is generated by x near x_{i-1} and u away from it. Similarly, $\mathcal{F}_v$ is generated by y near x_i and v away from it. By the previous paragraph, these local generators are a basis of $\mathcal{I}/\mathcal{I}^2$ at every point of R_i and we thus have $\mathcal{I}/\mathcal{I}^2 \cong \mathcal{F}_u \oplus \mathcal{F}_v$. We also have that $\mathcal{F}_u \cong \mathcal{O}_{R_i}(x_{i-1}) \cong \mathcal{O}_{\mathbb{P}^1}(1)$, since u vanishes to order one exactly at x_{i-1}. Symmetrically, $\mathcal{F}_v \cong \mathcal{O}_{R_i}(x_i) \cong \mathcal{O}_{\mathbb{P}^1}(1)$, since v vanishes to order one at x_i. Thus, after dualizing, we get $\mathcal{N}_{R_i/X} \cong \mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus 2}$. □

Suppose we have a semistable n-marked family of nodal curves $(C, \sigma_\bullet) \to Y$ as in Notation 2.4.4 with regular total space, and that additionally we have a line bundle L on C whose restriction to Y° is nef relative to the stabilization. Since C is regular, every node of $C_\mathfrak{o}$ is preserved over exactly one of Y_u and Y_v. Set $d_i := \deg(L|_{R_i})$ for $i = 1, \ldots, N$. By Lemma 2.4.5, the curve R_i may be flipped whenever one of x_{i-1}, x_i is preserved over Y_u and the other over Y_v, and (4) then gives

$$d'_{i-1} = d_{i-1} + d_i, \qquad d'_i = -d_i, \qquad d'_{i+1} = d_{i+1} + d_i. \tag{8}$$

We distinguish the two cases of Notation 2.4.4. Firstly, if z_R is a node, the moves of (8) can be done for $1 \le i \le N$. On the other hand, if z_R is a marking, Lemma 2.4.5 does not apply to R_N, and the moves of (8) can only be done for $1 \le i \le N - 1$. In both cases we add one further integer d_0 to the degree sequence which represents the degree of L on the rest of the fibre $C_\mathfrak{o}$. If z_R is a node, we work modulo $N + 1$, so that $d_{N+1} = d_0$.

Lemma 2.4.6. *If one starts with a sequence of integers $d_0, d_1, \ldots, d_N$ as in the situation above and applies any sequence of moves (8), where each move uses some $1 \le i \le N$ for which $d_i < 0$, then this process must terminate in a sequence with $d'_i \ge 0$ for all $1 \le i \le N$.*

Proof. Let $D_j := \sum_{i=0}^{j} d_i$ for $j = 0, \ldots, N$; we think of D_j as a label attached to the special point x_j. Since L is nef over Y°, the labels D_j at the nodes preserved over Y_u are weakly increasing in j, and similarly for the labels at the nodes preserved over Y_v. Moreover, if z_R is a marking, then $D_N \ge D_j$ for all j.

The condition that $d_i < 0$ is equivalent to $D_i < D_{i-1}$. For every $1 \le i \le N$, then (8) is equivalent to $D'_{i-1} = D_i$, $D'_i = D_{i-1}$, and $D'_j = D_j$ for all $j \ne i, i-1$. In other words, each move chooses a pair of adjacent elements (D_{i-1}, D_i) for $i \in 1, \ldots, N$ that are strictly decreasing and swaps them.

A move only exchanges two nodes with one preserved over Y_u and one over Y_v, and it never moves the marking. Thus, the sequence of D_j labels at the nodes preserved over Y_u is unchanged by a move, and similarly for Y_v and for the label of the marking. Thus every move preserves that the D_j at the nodes preserved over Y_u are weakly increasing in j, and similarly for the labels at the nodes preserved

over Y_v. Consequently, if $d_i < 0$ are some point, then x_{i-1} and x_i are preserved over different Y_u and Y_v, since otherwise $D_{i-1} \leq D_i$. Thus the move at i is allowed. If z_R is a marking in particular, $d_N \geq 0$ at every step.

Each moves strictly decreases the number of pairs $j < k$ where $D_j > D_k$, so this process terminates after finitely many steps. At that point, no d_i is negative for $1 \leq i \leq N$, so we have $D_0 \leq D_1 \leq \cdots \leq D_N$, which corresponds to $d_i = D_i - D_{i-1} \geq 0$ for $1 \leq i \leq N$. □

2.5 Θ-completeness and S-completeness

We first prove Θ- and S-completeness of the stack $\mathcal{X}^{\mathrm{rep}}_{g,n}$ under the assumption that the generic fibre is smooth, and then reduce the general case to this one. As in Notation 2.2.3, we use Y to denote either the surface $\mathrm{Spec}(R[s,t]/(st-\pi))$ or $\mathrm{Spec}(R[t])$, and $Y^\circ := Y \backslash \mathfrak{o}$ for a DVR R.

Theorem 2.5.1. *Fix a morphism $\mathcal{Y}^\circ \to \mathcal{X}^{\mathrm{rep}}_{g,n}$ with corresponding composition $Y^\circ \to \mathcal{Y}^\circ \to \mathcal{X}^{\mathrm{rep}}_{g,n}$. If the generic fibre of the corresponding family of curves $C^\circ \to Y^\circ$ is smooth, an extension $Y \to \mathcal{X}^{\mathrm{rep}}_{g,n}$ exists and is unique.*

Proof. Since $\mathcal{X}^{\mathrm{rep}}_{g,n}$ has affine diagonal, the filling is unique (see Remark 2.2.2), so we only need to show existence. Furthermore, after passing to an extension of the DVR, we may assume that R has algebraically closed residue field.

The morphism $\mathcal{Y}^\circ \to \mathcal{X}^{\mathrm{rep}}_{g,n}$ corresponds to a family $(C^\circ, \sigma^\circ_\bullet, L^\circ) \to \mathcal{Y}^\circ$ with smooth generic fibre. We construct an extension $(C, \sigma_\bullet, L) \to \mathcal{Y}$ of the triple $(C^\circ, \sigma^\circ_\bullet, L^\circ)$ so that L° is relatively ample for the stabilization morphism. Our proof proceeds in three steps.

Step 1. Let $(C'^{,\circ}, \sigma'^{,\circ}_\bullet) \to (C^\circ, \sigma^\circ_\bullet)$ be the minimal resolution as in the proof of Lemma 2.3.1, so that $C'^{,\circ}$ is regular. We denote by $L'^{,\circ}$ the pullback of the line bundle L° to $C'^{,\circ}$. Note that $L'^{,\circ}$ is nef relative to the stabilization. By Corollary 2.3.5, there is an extension of $(C'^{,\circ}, \sigma'^{,\circ}_\bullet)$ to a family $(C', \sigma'_\bullet)$ of semistable curves over $\mathcal{Y}$ such that C' is regular. Since the total space C' is regular, we can extend $L'^{,\circ}$ to a line bundle L' on C'. By construction, we have that the restriction of $(C', \sigma'_\bullet, L')$ to $\mathcal{Y}^\circ$ recovers $(C'^{,\circ}, \sigma'^{,\circ}_\bullet, L'^{,\circ})$. We abuse notation and denote by the same symbols the corresponding $\mathbb{G}_m$-equivariant families over the schemes Y and Y°, respectively.

Step 2. Using Lemma 2.4.3, flip the irreducible components $R \cong \mathbb{P}^1$ of rational bridges in the central fibre $C'_\mathfrak{o}$ such that $\deg(L'|_R) < 0$. After each flip we still get a family of semistable curves over Y. The rational curves R in the central fibre with $\deg(L'|_R) < 0$ all have one node preserved over Y_u, and one node preserved over Y_v, where we use the notation (u, v) of Notation 2.4.4 for the pair of parameters cutting out $\mathfrak{o} \in Y$. Thus by Lemma 2.4.5 the flops are all Atiyah flops. If both nodes are preserved over Y_u (or both over Y_v), we do not need to flip such R since $\deg(L'|_R) \geq 0$ by the fact that L' is nef on the rational bridges in the fibers of $C'^{,\circ}$ over Y°. We can always perform a sequence of flips until we get a nef line bundle by Lemma 2.4.6. Denote the family after flips still by $(C', \sigma'_\bullet, L')$.

Step 3. Let $(C^{\mathrm{st}}, \sigma^{\mathrm{st}}_\bullet)$ denote the stabilization of $(C', \sigma'_\bullet)$. By Lemma 2.4.1, L' is semiample relative to C^{st} and we get a map $(C', \sigma'_\bullet) \to (C, \sigma_\bullet)$ contracting rational curves R with $\deg(L'|_R) = 0$, where $(C, \sigma_\bullet)$ is a family of n-marked semistable curves over Y. Moreover, L' descends to a line bundle L on C which is ample relative to C^{st}; this follows directly from the relative Proj construction over C^{st} and the fact

that $C' \to C$ is given by a relative linear series of a power of L'. The restriction of $(C, \sigma_\bullet, L)$ to Y° recovers $(C^\circ, \sigma_\bullet^\circ, L^\circ)$. Indeed, by the relative ampleness of L°, the curve C° is the relative Proj of $\bigoplus_{\ell \geq 0} \mathrm{st}^\circ_*((L^\circ)^{\otimes \ell})$, where $\mathrm{st}^\circ : (C^\circ, \sigma_\bullet^\circ) \to (C^{\mathrm{st},\circ}, \sigma_\bullet^{\mathrm{st},\circ})$ is the stabilization morphism. □

We now explain how to fill families where the generic fibre is of genus zero. For the following result, we define $\mathcal{X}^{\mathrm{rep}}_{0,2}$ similarly as in Definition 2.1.4, with the convention that for condition (2) of Definition 1.0.1, we require L to be ample relative to the base T.

Proposition 2.5.2. *Let $f : \mathcal{Y}^\circ \to \mathcal{X}^{\mathrm{rep}}_{0,2}$ be a family $(C^\circ, \sigma_1^\circ, \sigma_2^\circ, \mathcal{L}^\circ)$ of 2-marked twisted curves of genus 0 such that the generic fiber of C° is isomorphic to $\mathbb{P}^1$. Then, there is a morphism $\widetilde{f} : Y \to \mathcal{X}^{\mathrm{rep}}_{0,2}$ extending the composition $Y^\circ \to \mathcal{Y}^\circ \to \mathcal{X}^{\mathrm{rep}}_{0,2}$.*

Proof. Let $C'^{,\circ} \to C^\circ$ be the minimal resolution as in the proof of Lemma 2.3.1, so that $C'^{,\circ}$ is regular, and let $L'^{,\circ}$ be the pullback of L° to $C'^{,\circ}$. Let $\sigma_1'^{,\circ}, \sigma_2'^{,\circ}$ denote the flat closures in $C'^{,\circ}$ of the generic fibers of the markings $\sigma_1^\circ, \sigma_2^\circ$. We have that $(C'^{,\circ}, \sigma_1'^{,\circ}, \sigma_2'^{,\circ})$ is a 2-marked family of genus 0 curves over $\mathcal{Y}^\circ$.

By Lemma 2.3.6, we get a 2-marked family $(C', \sigma_1', \sigma_2', L')$ over $\mathcal{Y}$, such that the restriction of $(C', \sigma_1', \sigma_2')$ to $\mathcal{Y}^\circ$ recovers $(C'^{,\circ}, \sigma_1'^{,\circ}, \sigma_2'^{,\circ})$. Also by Lemma 2.3.6, C' is regular, so the line bundle $L'^{,\circ}$ extends to a line bundle L' on C'. We abuse notation and denote by the same symbols the corresponding $\mathbb{G}_m$-equivariant families over the schemes Y and Y° respectively.

The filling for the 2-marked genus zero family on the level of coarse spaces can now be obtained in the same way as Theorem 2.5.1. That is, flip rational curves R in the central fibre where $\deg(L'|_R) < 0$ until L' is nef, and then contract rational curves where L' has degree zero using Lemma 2.4.1. This gives a contraction $C' \to C$ where $(C, \sigma_1, \sigma_2) \to Y$ is a family of nodal curves, and L' descends to a line bundle L on C. □

To prove Θ- and S-completeness of $\mathcal{X}^{\mathrm{rep}}_{g,n}$, we reduce to the case of Theorem 2.5.1 and Proposition 2.5.2 by taking the normalization of the family of curves and then gluing.

Theorem 2.5.3. *The stack $\mathcal{X}^{\mathrm{rep}}_{g,n}$ is Θ-complete and S-complete.*

Proof. Since $\mathcal{X}^{\mathrm{rep}}_{g,n}$ is open $\mathcal{X}_{g,n}$ by Proposition 2.1.5, it has affine diagonal. Thus we only need to show existence of the filling and we may replace R with an arbitrary extension of DVR's (see Remark 2.2.2). We start with a family $(C^\circ, \sigma_\bullet^\circ, L^\circ)$ over $\mathcal{Y}^\circ$. By Lemma 2.2.6, it suffices to extend to a family $(C, \sigma_\bullet, L)$ over Y.

Take the normalization $\nu : \widetilde{C}^\circ \to C^\circ$, and let $\Delta \subset \widetilde{C}^\circ$ denote the closed reduced subscheme underlying the preimage of the nodal locus of C° (i.e. the conductor). Then $\Delta \to \mathcal{Y}^\circ$ is étale, proper and quasifinite, and it is contained in the smooth locus of $\widetilde{C}^\circ \to \mathcal{Y}^\circ$. After passing to an extension of the DVR R, we may assume that the generic fibers of every connected component of both Δ and C° are geometrically connected. This implies that Δ is a disjoint union $\Delta = \bigsqcup_i \xi_i^\circ$ of some even number $2m$ of markings. Considering in addition the markings $\sigma_\bullet^\circ$ already present on C°, we view $(\widetilde{C}^\circ, \sigma_\bullet^\circ, \xi_\bullet^\circ)$ as a family of disconnected curves with $n + 2m$ markings indexed by the set $[n] \sqcup [2m]$. We denote by $\mathrm{inv} : [2m] \to [2m]$ the unique fixed-point free involution whose orbits are the pairs of markings ξ_i° with the same image in the nodal locus of C°. We have identifications $\xi_i^\circ \xrightarrow{\sim} \xi_{\mathrm{inv}(i)}^\circ$ for all $i \in [2m]$ and the pushout of $\widetilde{C}^\circ$ obtained by gluing these pairs recovers C°.

Let $\widetilde{C}^\circ = \bigsqcup_{i=1}^{\ell} \widetilde{C}_i^\circ$ be the decomposition into connected components, so that each $\widetilde{C}_i^\circ$ is a $\mathbb{G}_m$-equivariant family of curves over Y° with smooth generic fiber. For each i, let g_i denote the genus of the fibers of $\widetilde{C}_i^\circ$, and n_i (resp. m_i) denote the number of markings $\sigma_\bullet^\circ$ (resp. $\xi_\bullet^\circ$) lying on $\widetilde{C}_i^\circ$. We may then think of $\left(\widetilde{C}_i^\circ, \sigma_\bullet^\circ \cap \widetilde{C}_i^\circ, \xi_\bullet^\circ \cap \widetilde{C}_i^\circ, \nu^*(L^\circ)|_{\widetilde{C}_i^\circ}\right)$ as a family $\mathcal{Y}^\circ \to \mathcal{X}^{\mathrm{rep}}_{g_i, n_i+m_i}$ with smooth generic fiber.

We extend this to a family $(\widetilde{C}_i, \sigma_\bullet \cap \widetilde{C}_i, \xi_\bullet \cap \widetilde{C}_i, \widetilde{L}|_{\widetilde{C}_i})$ in $\mathcal{X}^{\mathrm{rep}}_{g_i, n_i+m_i}$ over Y. Indeed, if $g_i \geq 2$, we use Theorem 2.5.1. If $g_i = 1$, then by construction the number of markings $n_i + m_i$ is at least 1, and so we have access to the stabilization morphism. The same argument as in Theorem 2.5.1 then proves the existence of the extension. Finally, if $g_i = 0$, then $n_i + m_i = 2$, and so we may apply Proposition 2.5.2. In summary, we have extended the $(n+2m)$-marked (disconnected) family $(\widetilde{C}^\circ, \sigma_\bullet^\circ, \xi_\bullet^\circ, \nu^*(L^\circ))$ to a family $(\widetilde{C}, \sigma_\bullet, \xi_\bullet, \widetilde{L})$ over Y.

We note that for each $i \in [2m]$, the automorphism $\xi_i^\circ \xrightarrow{\sim} \xi_{\mathrm{inv}(i)}^\circ$ extends uniquely to $\xi_i \xrightarrow{\sim} \xi_{\mathrm{inv}(i)}$ over Y by Hartogs's theorem. By the existence of pushouts in [AHHLR24, Thm. 1.8], we may glue these pairs of subschemes $\xi_\bullet$ to obtain a family $(C, \sigma_\bullet)$ of n-marked curves extending $(C^\circ, \sigma_\bullet^\circ)$. Furthermore, since $\nu^*(L)$ is pulled back from C°, for each $i \in [2m]$ we obtain an identification $\nu^*(L)|_{\xi_i^\circ} \xrightarrow{\sim} \nu^*(L)|_{\xi_{\mathrm{inv}(i)}^\circ}$, where we are using the identifications $\xi_i^\circ \xrightarrow{\sim} \xi_{\mathrm{inv}(i)}^\circ$ to view these as line bundles on the same stack. By Hartogs' theorem, this extends uniquely to an identification $\widetilde{L}|_{\xi_i} \xrightarrow{\sim} \widetilde{L}|_{\xi_{\mathrm{inv}(i)}}$. Since the gluing [AHHLR24, Thm. 1.8] constructed above is a pushout in the category of quasi-separated stacks, by considering the target stack $B\mathbb{G}_m$ we see that these identifications are exactly the data needed to glue the line bundle $\widetilde{L}$ to a line bundle L on C extending L°. This yields the desired extension $(C, \sigma_\bullet, L)$ over Y. □

3 Relatively polarized curves

In this section we analyze the stack of relatively polarized curves $\mathrm{P}\overline{\mathcal{M}}_g$ introduced in Definition 1.0.4, and we prove the main result, Theorem 1.0.7.

3.1 General analysis of relatively polarized curves

Suppose $(\mathcal{C}, \mathcal{L}, s)$ is a degree 0 anchored curve over an algebraically closed field, and $\mathcal{R} \subset \mathcal{C}$ is a rational bridge. We let $\widetilde{\mathcal{C}} := \overline{(\mathcal{C} \setminus \mathcal{R})} \subset \mathcal{C}$ be the twisted curve whose coarse space is the partial normalization of C^{st} obtained by separating the node over which $\mathcal{R}$ lies.

Recall from the introduction the following terminology: An *orientation* of $\mathcal{R}$ is a choice of labels (p_-, p_+) for the two points $\widetilde{\mathcal{C}} \cap \mathcal{R}$. Given an orientation of $\mathcal{R}$, we let $\mathrm{ord}_\pm(\mathcal{R})$ denote the order of vanishing of the section $s^\vee : \mathcal{O}_{\widetilde{\mathcal{C}}} \to \mathcal{L}^\vee|_{\widetilde{\mathcal{C}}}$ in an étale neighborhood of $p_\pm$ divided by the order of $\mathrm{Aut}(p_\pm)$. A *relatively polarized curve* is a degree 0 anchored curve $(\mathcal{C}, \mathcal{L}, s)$ such that $\mathrm{ord}_\pm(\mathcal{R}) \leq 1$ and $\mathrm{ord}_+(\mathcal{R}) + \mathrm{ord}_-(\mathcal{R}) = \deg(\mathcal{L}|_{\mathcal{R}})$ for every rational bridge $\mathcal{R} \subset \mathcal{C}$. We denote by $\mathrm{P}\overline{\mathcal{M}}_g \subset \mathcal{A}nc_g$ the substack of relatively polarized curves.

Lemma 3.1.1. *For any geometric point $(\mathcal{C}, \mathcal{L}, s)$ in $\mathrm{P}\overline{\mathcal{M}}_g$, any vanishing point of the cosection s and any orbifold point of $\mathcal{C}$ must lie on a rational bridge of $\mathcal{C}$.*

Proof. The condition $\mathrm{ord}_+(\mathcal{R}) + \mathrm{ord}_-(\mathcal{R}) = \deg(\mathcal{L}|_{\mathcal{R}})$ implies that if s vanished at any point that was not on a rational bridge, then the total degree of $\mathcal{L}^\vee$ on all of the

bridges would be less than the total degree of $\mathcal{L}^\vee$ on all of the stable components of $\mathcal{C}$, which would contradict the fact that the total degree of $\mathcal{L}$ is zero. As a consequence, s trivializes $\mathcal{L}$ at any node x that is not on a rational bridge, which means that $\mathrm{Aut}(x)$ must be trivial, because it acts faithfully on the fiber $\mathcal{L}_x$. □

Definition 3.1.2 (Type of a rational bridge)**.** Given a geometric point $(\mathcal{C}, \mathcal{L}, s)$ of $\mathrm{P}\overline{\mathcal{M}}_g$ and an oriented rational bridge $\mathcal{R} \subset \mathcal{C}$, order the components of $\mathcal{R} = \mathcal{S}_1 \cup \cdots \cup \mathcal{S}_m$ such that only adjacent $\mathcal{S}_i$ meet and $p_- \in \mathcal{S}_1$. We define the subset $\mathrm{type}(\mathcal{R}) \subset [-1,1] \cap \mathbb{Q}$ as the set $\{0, \deg(\mathcal{L}|_{\mathcal{S}_1}), \deg(\mathcal{L}|_{\mathcal{S}_1 \cup \mathcal{S}_2}), \ldots, \deg(\mathcal{L}|_{\mathcal{R}})\} - \mathrm{ord}_-(\mathcal{R})$.

Note that $\max(\mathrm{type}(\mathcal{R})) > 0 > \min(\mathrm{type}(\mathcal{R}))$ for any bridge $\mathcal{R}$.

Lemma 3.1.3. *Reversing the orientation of $\mathcal{R}$ has the effect of replacing $\mathrm{type}(\mathcal{R}) \mapsto -\mathrm{type}(\mathcal{R})$. So, for an unoriented rational bridge, $\mathrm{type}(\mathcal{R})$ is defined up to flipping signs.*

Proof. The condition that $\mathrm{ord}_-(\mathcal{R}) + \mathrm{ord}_+(\mathcal{R}) = \deg(\mathcal{L}|_{\mathcal{R}})$ implies that the definition above is equivalent to $\mathrm{type}(\mathcal{R}) = \{-\deg(\mathcal{L}|_{\mathcal{R}}), -\deg(\mathcal{L}|_{\mathcal{S}_2 \cup \cdots \cup \mathcal{S}_m}), \ldots, -\deg(\mathcal{L}|_{\mathcal{S}_m}), 0\} + \mathrm{ord}_+(\mathcal{R})$. □

Lemma 3.1.4. *Let $(\mathcal{C}, \mathcal{L}, s)$ be a geometric point of $\mathrm{P}\overline{\mathcal{M}}_g$. Let $\{\mathcal{R}_i\}_{i=1}^{\ell}$ be collection of bridges of $\mathcal{C}$, and choose an orientation $(p_-^{(i)}, p_+^{(i)})$ for each $\mathcal{R}_i$. Then, the following hold:*

(1) The restriction of $\mathcal{L}^\vee$ to the union $\widetilde{\mathcal{C}}^{st} = \overline{\mathcal{C} \setminus \sqcup_i \mathcal{R}_i}$ of stable components is isomorphic to $\mathcal{O}(\mathcal{D})$ for the following Cartier divisor

$$\mathcal{D} := \sum_{i=1}^{\ell} \left(\mathrm{ord}_-(\mathcal{R}_i) p_-^{(i)} + \mathrm{ord}_+(\mathcal{R}_i) p_+^{(i)} \right).$$

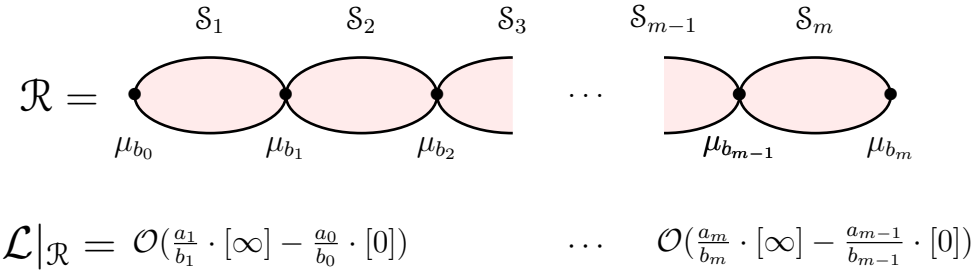
(2) The cosection s is given as the composition

$$\mathcal{L} \twoheadrightarrow \mathcal{L}|_{\widetilde{\mathcal{C}}^{\mathrm{st}}} \xrightarrow{\psi} \mathcal{O}_{\widetilde{\mathcal{C}}^{st}}(-\mathcal{D}) \hookrightarrow \mathcal{O}_{\mathcal{C}}$$

for some choice of isomorphism $\psi : \mathcal{L}|_{\widetilde{\mathcal{C}}^{\mathrm{st}}} \xrightarrow{\sim} \mathcal{O}_{\widetilde{\mathcal{C}}^{st}}(-\mathcal{D})$.

(3) Choose a bridge $\mathcal{R} \subset \mathcal{C}$ and consider the tuple $\mathrm{type}(\mathcal{R}) = \left(\frac{a_j}{b_j}\right)_{j=0}^{m}$, where we express each term in reduced fraction form. Then the jth irreducible component $\mathcal{S}_j$ is isomorphic to the stack $\mathbb{P}^1 \left[\frac{1}{b_{j-1}} \mid \frac{1}{b_j}\right]$ obtained from $\mathbb{P}^1$ by taking the root b_{j-1}^{th}-root stack at 0 and the b_j^{th}-root stack at ∞. Under this identification, we have $\mathcal{L}|_{\mathcal{S}_j} \cong \mathcal{O}(\frac{a_j}{b_j} \cdot [\infty] - \frac{a_{j-1}}{b_{j-1}} \cdot [0])$. In other words, the bridge $\mathcal{R}$ and line bundle $\mathcal{L}|_{\mathcal{R}}$ are as pictured in Figure 1 below.

Figure 1:



Proof. Parts (1) and (2) are direct consequences of Lemma 3.1.1 and the definition of $\mathrm{ord}_-(\mathcal{R}_i)$ and $\mathrm{ord}_+(\mathcal{R}_i)$ as orders of vanishing of the cosection at the endpoints of the bridge $\mathcal{R}_i$, using that the relative ampleness of $\mathcal{L}|_{\mathcal{R}_i}$ forces the cosection to vanish on every rational bridge $\mathcal{R}_i$.

For Part (3), we note that we must have $\mathcal{S}_j \cong \mathbb{P}^1\left[\frac{1}{d_{j-1}} \mid \frac{1}{d_j}\right]$ for some tuple of positive integers $(d_j)_{j=0}^m$. We use freely the description of the Picard group of $\mathbb{P}^1\left[\frac{1}{d_{j-1}} \mid \frac{1}{d_j}\right]$ given in [MT15, Prop. 2.2], and the fact that it is generated by the line bundles $\mathcal{O}(\frac{1}{d_{j-1}} \cdot [0])$ and $\mathcal{O}(\frac{1}{d_j} \cdot [\infty])$. By the definition of $\mathrm{ord}_-(\mathcal{R}_i)$ and the fact that the isotropy group of p_-^i acts faithfully on the fiber of $\mathcal{L}$, it follows the isotropy group of $p_-^i = 0 \in \mathcal{S}_1$ is μ_{b_0} and the corresponding character of the fiber of $\mathcal{L}$ is the same as for $\mathcal{O}(-\frac{a_0}{b_0} \cdot [0])$. Since the degree of $\mathcal{L}|_{\mathcal{S}_1}$ is $\frac{a_1}{b_1} - \frac{a_0}{b_0}$ and the isotropy group of $\infty \in \mathcal{S}_1$ acts faithfully on the fiber of $\mathcal{L}$, this forces the isotropy group of ∞ to be μ_{b_1} and $\mathcal{L}|_{\mathcal{S}_1} \cong \mathcal{O}(\frac{a_1}{b_1} \cdot [\infty] - \frac{a_0}{b_0} \cdot [0])$. It follows that the isotropy group of $0 \in \mathcal{S}_2$ is μ_{b_1} and the corresponding character of the fiber of $\mathcal{L}$ agrees with the one for $\mathcal{O}(-\frac{a_1}{b_1} \cdot [0])$. Using that the degree of $\mathcal{L}|_{\mathcal{S}_2}$ is $\frac{a_2}{b_2} - \frac{a_1}{b_1}$, we conclude that $\mathcal{S}_2 \cong \mathbb{P}^1\left[\frac{1}{b_1} \mid \frac{1}{b_2}\right]$ and $\mathcal{L}|_{\mathcal{S}_2} \cong \mathcal{O}(\frac{a_2}{b_2} \cdot [\infty] - \frac{a_1}{b_1} \cdot [0])$. Iterating this procedure yields the desired result for all components $\mathcal{S}_j$. □

Lemma 3.1.5. *Let $(\mathcal{C}, \mathcal{L}, s)$ be a geometric point of $\mathcal{A}nc_g$. Then, for all integers n we have $\chi(\mathcal{L}^{\otimes n}) = n\deg(\mathcal{L}) + 1 - g$.*

Proof. The desired equality holds for sufficiently divisible n, by the definition of degree and an application of Riemann-Roch to the descended powers of the line bundle on the coarse space. Hence, it suffices to show that the function $n \mapsto \chi(\mathcal{L}^{\otimes n})$ is linear.

Let $\{j_i : \mathcal{G}_{p_i} \hookrightarrow \mathcal{C}\}_{i=1}^{\ell}$ be residual gerbes of all nodes p_i of $\mathcal{C}$. Let $\nu : \widetilde{\mathcal{C}} \to \mathcal{C}$ be the normalization, and let x_i, y_i be the pair of preimages of p_i. Let us denote by $\{\widetilde{\mathcal{C}}_q\}_q$ the set of connected components of $\widetilde{\mathcal{C}}$, with corresponding closed immersions $f_q : \widetilde{\mathcal{C}}_q \hookrightarrow \mathcal{C}$. We have a short exact sequence of sheaves for all integers n

$$0 \to \mathcal{L}^{\otimes n} \to \bigoplus_q (f_q)_* f_q^* \mathcal{L}^{\otimes n} \to \bigoplus_i (j_i)_* j_i^* \mathcal{L}^{\otimes n} \to 0. \tag{9}$$

By looking at the characters of isotropy groups of p_i acting on $\mathcal{L}$, we may define a unique collection of rational numbers $\{a_i\}_{i=1}^{\ell}$ with $0 \le a_i < 1$ such that for all q the line bundle

$$\mathcal{M}_q = \mathcal{O}\left(\sum_{i=1}^{\ell} \left(a_i \cdot [x_i \cap \widetilde{\mathcal{C}}_q] - a_i \cdot [y_i \cap \widetilde{\mathcal{C}}_q]\right)\right) \in \mathrm{Pic}(\widetilde{\mathcal{C}}_q)$$

satisfies that the isotropy group acts trivially on $\mathcal{N}_q := \mathcal{L}|_{\widetilde{\mathcal{C}}_q} \otimes \mathcal{M}_q^{\vee}$. Replacing $\mathcal{L}|_{\widetilde{C}_q} = \mathcal{N}_q \otimes \mathcal{M}_q$ in (9) and using the fact that $\mathcal{N}_q$ is trivializable when restricted to the residual gerbes at any point, we get a short exact sequence

$$0 \to \mathcal{L}^{\otimes n} \to \bigoplus_q (f_q)_* (\mathcal{N}_q^{\otimes n} \otimes \mathcal{M}_q^{\otimes n}) \to \bigoplus_i (j_i)_* (\mathcal{M}_q^{\otimes n}|_{\mathcal{G}_{x_i}}) \to 0.$$

Taking Euler characteristics, we get

$$\chi(\mathcal{L}^{\otimes n}) = \sum_q \chi(\mathcal{N}_q^{\otimes n} \otimes \mathcal{M}_q^{\otimes n}) - \sum_i \chi(\mathcal{M}_q^{\otimes n}|_{\mathcal{G}_{x_i}}) \tag{10}$$

Note that

$$\chi(\mathcal{M}_q^{\otimes n}|_{\mathcal{G}_{x_i}}) = h(na_i), \tag{11}$$

where $h : \mathbb{Q} \to \{0, 1\}$ is the indicator function of the integers $\mathbb{Z} \subset \mathbb{Q}$. On the other hand, if we let $\pi_q : \widetilde{\mathcal{C}}_q \to \widetilde{C}_q$ denote the coarse space morphism, then we have by the projection formula that

$$\chi(\mathcal{N}_q^{\otimes n} \otimes \mathcal{M}_q^{\otimes n}) = \chi(N_q^{\otimes n} \otimes (\pi_q)_*(\mathcal{M}_q^{\otimes n})),$$

where N_q is the descended line bundle with $(\pi_q)^*(N_q) = \mathcal{N}_q$. Furthermore, if we also denote by x_i and y_i the corresponding images in the coarse space, then we have that $(\pi_q)_*(\mathcal{M}_q^{\otimes n})$ is the line bundle $\mathcal{O}\left(\lfloor na_i \rfloor \cdot [x_i \cap \widetilde{C}_q] - \lceil na_i \rceil \cdot [y_i \cap \widetilde{C}_q]\right)$. If we denote by g_q the genus of the smooth curve $\widetilde{C}_q$, then by Riemann-Roch, adding over all q and using that each point x_i, y_i appears only once in the normalization $\widetilde{\mathcal{C}}$, we conclude that

$$\sum_q \chi(N_q^{\otimes n} \otimes (\pi_q)_*(\mathcal{M}_q^{\otimes n})) = \sum_q n \deg(N_q) + \sum_{i=1}^{\ell} (\lfloor na_i \rfloor - \lceil na_i \rceil) + \sum_q (1 - g_q).$$

Using the expression $\lfloor na_i \rfloor - \lceil na_i \rceil = h(na_i) - 1$ in terms of the indicator function as above, we may rewrite this as

$$\sum_q \chi(N_q^{\otimes n} \otimes (\pi_q)_*(\mathcal{M}_q^{\otimes n})) = \sum_q n \deg(N_q) + \sum_{i=1}^{\ell} (h(na_i) - 1) + \sum_q (1 - g_q). \tag{12}$$

Combining equations (10), (11) and (12), we conclude that

$$\chi(\mathcal{L}^{\otimes n}) = \sum_q n \deg(N_q) - \ell + \sum_q (1 - g_q),$$

which is linear as a function of n. □

Remark 3.1.6. The proof of statement Lemma 3.1.5 applies in a more general setting: an unmarked twisted curve $\mathcal{C}$ of genus g equipped with a line bundle $\mathcal{L}$ satisfies $\chi(\mathcal{L}^{\otimes n}) = n \deg(\mathcal{L}) + (1 - g)$ for all integers n.

Rational models for nodes

Let $P^{\mathrm{un}} \to \mathbb{A}^1$ denote the versal deformation of the nodal curve $\mathbb{P}^1 \cup \mathbb{P}^1$, i.e., it is an isotrivial degeneration of one line to two lines meeting at a node $x \in P^{\mathrm{un}}$. The family $P^{\mathrm{un}} \to \mathbb{A}^1$ can be realized as the composition of the blow-up $B \to \mathbb{P}^1 \times \mathbb{A}^1$ at a point, with the second projection $\mathbb{P}^1 \times \mathbb{A}^1 \to \mathbb{A}^1$. We fix an orientation of the node x by writing the zero fiber as $P_0^{\mathrm{un}} = P_0^{\mathrm{un},-} \cup P_0^{\mathrm{un},+}$.

The versality of the family $P^{\mathrm{un}} \to \mathbb{A}^1$ means the following: For any family of nodal curves $C \to \mathrm{Spec}(R)$ for a complete local Noetherian algebra R with residue field k, if $x \in C(k)$ is a node lying over the closed point, then there is a morphism $\mathrm{Spec}(R) \to \mathbb{A}^1$ taking the closed point to 0 and an isomorphism $\varphi : \widehat{C} \cong \widehat{P_R}$ between the formal completion of C at x and the formal completion of $P_R := \mathrm{Spec}(R) \times_{\mathbb{A}^1} P^{\mathrm{un}}$ at the unique node over the closed point of $\mathrm{Spec}(R)$ [Sta25, Tag 0CBX]. We call this data a *rational model* around the node $x \in C(k)$. The preimage of the node in P_0^{un} is a closed subscheme $Z \hookrightarrow P_R$ such that composing with the projection to

$\mathrm{Spec}(R)$ is also a closed immersion $Z \hookrightarrow \mathrm{Spec}(R)$, hence Z is complete, and therefore φ identifies Z with a closed subscheme of C as well.

The following lemma applies to a complete local Noetherian algebra R with residue field k, a family of twisted curves $\mathcal{C} \to Y$ over a Noetherian R-scheme Y, an invertible sheaf $\mathcal{L}$ on $\mathcal{C}$, and a cosection $s : \mathcal{L} \to \mathcal{O}_{\mathcal{C}}$. We suppose that there is a stable family of nodal curves $C^{\mathrm{st}} \to \mathrm{Spec}(R)$ and a morphism $\pi : \mathcal{C} \to C^{\mathrm{st}}$ such that the morphism $\mathcal{C} \to Y \times_{\mathrm{Spec}(R)} C^{\mathrm{st}}$ is the stabilization in every fiber over Y.

Lemma 3.1.7. *Let $x \in C^{\mathrm{st}}(k)$ be a node, and let $\mathrm{Spec}(R) \to \mathbb{A}^1$ and $\varphi : \widehat{C^{\mathrm{st}}} \cong \widehat{P_R}$ be a rational model around x. Assume that $V(s) \setminus \pi^{-1}(Z) \subset |\mathcal{C}|$ is closed. Then the pushout*

$$\begin{array}{ccc} \mathcal{C} \times_{C^{\mathrm{st}}} (\mathrm{Spec}(\widehat{\mathcal{O}}_{C^{\mathrm{st}},x}) \setminus Z) & \xrightarrow{\varphi} & Y \times_{\mathrm{Spec}(R)} (P_R \setminus Z) \\ \downarrow & & \downarrow \\ \mathcal{C} \times_{C^{\mathrm{st}}} \mathrm{Spec}(\widehat{\mathcal{O}}_{C^{\mathrm{st}},x}) & \longrightarrow & \mathcal{P} \end{array}$$

exists in the category of algebraic stacks with affine diagonal, and $\mathcal{P}$ is a family of twisted chains of rational curves over Y. Furthermore:

- *$\mathcal{L}$ and s glue uniquely with $(s = 1 : \mathcal{O}_{P_R \setminus Z} \cong \mathcal{O}_{P_R \setminus Z})$ to obtain a line bundle $\mathcal{L}_{\mathcal{P}}$ and cosection $s_{\mathcal{P}} : \mathcal{L}_{\mathcal{P}} \to \mathcal{O}_{\mathcal{P}}$ on $\mathcal{P}$.*
- *For every $y \in Y$, the choice of orientation for P_0^{un} induces an orientation for the rational bridge $\mathcal{R}_y := \mathcal{C}_y \cap \pi^{-1}(Z) \subset \mathcal{C}_y$ and picks out a distinguished end component of the fiber $\mathcal{P}_y^- \subset \mathcal{P}_y$, and one has*

$$\mathrm{ord}_{\pm}(\mathcal{R}_y) = -\deg(\mathcal{L}_{\mathcal{P}}|_{\mathcal{P}_y^{\pm}}) \text{ and}$$

$$\mathrm{type}(\mathcal{R}_y) = \left\{ \deg(\mathcal{L}_{\mathcal{P}}|_{\mathcal{S}}) \mid \text{connected proper subcurves } \mathcal{S} \subsetneq \mathcal{P}_y \text{ that contain } \mathcal{P}_y^- \right\}.$$

Proof. The existence of the pushout is **??**, and it is a twisted curve because the pushout description itself can be used to give an étale local identification of $\mathcal{P}$ with families of twisted curves over Y. The existence and uniqueness of the gluing $s_{\mathcal{P}} : \mathcal{L}_{\mathcal{P}} \to \mathcal{O}_{\mathcal{P}}$ follows from the universal property of pushouts, applied to maps to $\mathbb{A}^1/\mathbb{G}_m$ – the key fact is that s is nowhere vanishing when restricted to the upper-left corner of the pushout diagram. We have $\mathrm{ord}_-(\mathcal{R}_y) = -\deg(\mathcal{L}_{\mathcal{P}}|_{\mathcal{P}_y^-})$, because if a section of a line bundle on an orbifold curve vanishes at a single point, then the orbifold notion of order of vanishing agrees with the orbifold notion of degree for the line bundle. The final expression for $\mathrm{type}(\mathcal{R}_y)$ results from the fact that $\mathcal{P}_y \cong \mathcal{P}_y^- \cup \mathcal{R}_y \cup \mathcal{P}_y^+$ as polarized chains. □

The new interpretation of $\mathrm{type}(\mathcal{R}_y)$ in terms of degrees of $\mathcal{L}$ on the rational model is the key idea leading to all of the results below.

3.2 Openness

Lemma 3.2.1. *Let $(\mathcal{C}, \mathcal{L}, s)$ be an R-point of $\mathcal{A}nc_g$, where R is a complete DVR. If the special fiber lies in $\mathrm{P}\overline{\mathcal{M}}_g$, then the whole family lies in $\mathrm{P}\overline{\mathcal{M}}_g$.*

Proof. We denote by η and 0 the generic and special points of $\mathrm{Spec}(R)$, respectively. Let $\mathcal{R}_\eta \subset \mathcal{C}_\eta$ be a rational bridge, and let $\mathcal{S} := \overline{\mathcal{R}_\eta} \cap \mathcal{C}_0 \subset \mathcal{C}_0$ be its flat limit. Let $\mathcal{R}_0 \subset \mathcal{C}_0$ be the rational bridge that contains $\mathcal{S}$, oriented compatibly with $\mathcal{R}_\eta$. It decomposes as $\mathcal{R}_0 = \mathcal{R}_0^- \cup \mathcal{S} \cup \mathcal{R}_0^+$. Let $D_\pm \subset V(s_\eta) \setminus \mathcal{R}_\eta \subset \mathcal{C}_\eta$ be the part of the vanishing locus whose limit in $\mathcal{C}_0$ lies in $\mathcal{R}_0^\pm \setminus \mathcal{S}$. This part of the vanishing locus must

be a collection of isolated points lying over the smooth locus of the stabilization of $\mathcal{C}$, so we may regard D_- and D_+ as effective divisors on $\mathcal{C}_\eta$. The lemma follows from the following key formula

$$\operatorname{ord}_\pm(\mathcal{R}_\eta) = \operatorname{ord}_\pm(\mathcal{R}_0) - \deg(\mathcal{L}|_{\mathcal{R}_0^\pm}) - \deg(D_\pm). \tag{13}$$

To see why this implies the lemma, the hypothesis that the special fiber lies in $\mathrm{P}\overline{\mathcal{M}}_g$ implies that $\operatorname{ord}_\pm(\mathcal{R}_0) \leq 1$. Because $\operatorname{ord}_\pm(\mathcal{R}_\eta) \geq 0$ and $\deg(D_\pm) \in \mathbb{Z}_{\geq 0}$, this implies that $\deg(D_\pm) = 0$, i.e., $D_\pm$ is empty – this is the main point where the restriction $\operatorname{ord}_\pm(\mathcal{R}) \leq 1$ in Definition 1.0.4 is needed. (13) also implies that $\operatorname{ord}_\pm(\mathcal{R}_\eta) \leq \operatorname{ord}_\pm(\mathcal{R}_0) \leq 1$. Finally, adding both versions of (13), and using the equality $\deg(\mathcal{L}|_{\mathcal{R}_\eta}) = \deg(\mathcal{L}|_{\mathcal{S}})$, we have

$$\operatorname{ord}_-(\mathcal{R}_\eta) + \operatorname{ord}_+(\mathcal{R}_\eta) - \deg(\mathcal{L}|_{\mathcal{R}_\eta}) = \operatorname{ord}_-(\mathcal{R}_0) + \operatorname{ord}_+(\mathcal{R}_0) - \deg(\mathcal{L}|_{\mathcal{R}_0^- \cup \mathcal{S} \cup \mathcal{R}_0^+}) = 0.$$

Applying these two conclusions to every rational bridge in $\mathcal{C}_\eta$ shows that the generic fiber lies in $\mathrm{P}\overline{\mathcal{M}}_g \subset \mathcal{A}nc_g$.

To prove (13), we can choose another family of twisted curves $\mathcal{C}' \to \operatorname{Spec}(R)$ and a map $\mathcal{C}' \to \mathcal{C}$ that is an isomorphism over the generic fiber and contracts rational subchains in the special fiber, and such that the closure of $D_\pm$ meets the smooth locus of $\mathcal{C}'_0$. It suffices to prove (13) for the pullback of $s : \mathcal{L} \to \mathcal{O}_\mathcal{C}$ to $\mathcal{C}'$, so we replace $\mathcal{C}$ with $\mathcal{C}'$ and assume the closure of $D_\pm$ lies in the smooth locus of $\mathcal{R}_0^\pm$.

The fact that s vanishes along D to the correct orders implies that we can factor $s = s' \cdot f$ for the canonical $f : \mathcal{O}_\mathcal{C}(-\overline{D}_\pm) \to \mathcal{O}_\mathcal{C}$. By construction, $s' : \mathcal{L}(\overline{D}_\pm) \to \mathcal{O}_\mathcal{C}$ is a cosection that does not vanish at any point in the generic fiber whose limit lies in $\mathcal{R}_0$. We may therefore apply Lemma 3.1.7 to the family $(\mathcal{C}, \mathcal{L}(\overline{D}_\pm), s')$ over $Y = \operatorname{Spec}(R)$, to produce an oriented family of rational chains $\mathcal{P} \to \operatorname{Spec}(R)$ that models the degeneration of bridges. Also, because the cosection f is nonvanishing at the endpoints of $\mathcal{R}_0$ and $\mathcal{R}_\eta$, the value of $\operatorname{ord}_\pm$ is the same for this new family of curves. The closure of the end components $\mathcal{P}_\eta^\pm$ of the generic fiber is a flat degeneration of $\mathcal{P}_\eta^\pm$ to $\mathcal{R}_0^\pm \cup \mathcal{P}_0^\pm$. The equation (13) then follows from the fact that $\deg(\mathcal{L}(\overline{D_\pm})|_{\mathcal{P}_\eta^\pm}) = -\operatorname{ord}_\pm(\mathcal{R}_\eta)$ and $\deg(\mathcal{L}(\overline{D_\pm})|_{\mathcal{P}_0^\pm}) = -\operatorname{ord}_\pm(\mathcal{R}_0)$. □

Lemma 3.2.2. *Suppose $(\mathcal{C}, \mathcal{L}, s)$ is an R point of $\mathrm{P}\overline{\mathcal{M}}_g$ for a complete DVR R. If $\mathcal{R}_\eta \subset \mathcal{C}_\eta$ is a rational bridge in the generic fiber, and $\mathcal{R}_0 \subset \mathcal{C}_0$ is the unique rational bridge in the special fiber that contains $\overline{R_\eta} \cap \mathcal{C}_0$, then for the orientation of $\mathcal{R}_0$ induced by an orientation of $\mathcal{R}_\eta$,*

$$\operatorname{type}(R_\eta) \subset \operatorname{type}(R_0).$$

Proof. After replacing R with a finite extension we may assume that the stabilization of the family $C^{\mathrm{st}} \to \operatorname{Spec}(R)$ has a section $x \in C^{\mathrm{st}}(R)$ whose generic fiber x_η is the node over which R_η lies. Then $\mathcal{R}_0$ is the preimage of the node x_0 in C_0^{st} and $\overline{\mathcal{R}_\eta} \cap \mathcal{C}_0 \subset \mathcal{R}_0$ is a subchain of rational curves.

We choose a rational model for the node $x_0 \in C^{\mathrm{st}}$ over $\operatorname{Spec}(R)$ compatible with the chosen orientation of $\mathcal{R}_\eta$. Lemma 3.1.7 applied with $Y = \operatorname{Spec}(R)$ gives a family of chains of rational orbifold curves $\mathcal{P} \to \operatorname{Spec}(R)$ such that for either point $z \in \operatorname{Spec}(R)$ the set $\operatorname{type}(\mathcal{R}_z) \subset \mathbb{Q}$ is exactly the set of degrees of connected proper subchains of $\mathcal{P}_z$ that contain $\mathcal{P}_z^-$. The flat limit of any subchain of this form in $\mathcal{P}_\eta$ is a subchain of the same form in $\mathcal{P}_0$, hence $\operatorname{type}(\mathcal{R}_\eta) \subset \operatorname{type}(\mathcal{R}_0)$. □

Proposition 3.2.3. *For all $\mathcal{B}$ as in Hypothesis 1.0.5, the substack $\mathrm{P}\overline{\mathcal{M}}_g^{\mathcal{B}} \subset \mathrm{P}\overline{\mathcal{M}}_g$ is open. In particular, $\mathrm{P}\overline{\mathcal{M}}_g \subset \mathcal{A}nc_g$ is open.*

Proof. Lemma 3.2.1 and Lemma 3.2.2 establish that $\mathrm{P}\overline{\mathcal{M}}_g^{\mathcal{B}} \subset \mathcal{A}nc_g$ is closed under generization, so to show that it is open, it suffices to show that $|\mathrm{P}\overline{\mathcal{M}}_g^{\mathcal{B}}| \subset |\mathcal{A}nc_g|$ is a locally constructible subset. We use the natural gluing morphisms

$$\mathcal{A}nc_{g_1,1} \times_{\mathbb{A}^1/\mathbb{G}_m} \mathcal{X}_{0,2} \times_{\mathbb{A}^1/\mathbb{G}_m} \mathcal{A}nc_{g_2,1} \to \mathcal{A}nc_{g_1+g_2} \text{ and}$$
$$\mathcal{A}nc_{g,2} \times_{(\mathbb{A}^1/\mathbb{G}_m)^2} \mathcal{X}_{0,2} \to \mathcal{A}nc_{g+1},$$

where $\mathcal{X}_{0,2}$ denotes the stack of marked rational (twisted) chains, regarded as a stack over $(\mathbb{A}^1/\mathbb{G}_m)^2$ by the morphism $\mathcal{X}_{0,2} \to (B\mathbb{G}_m)^2$ taking a polarized marked chain to the line bundle at its endpoints composed with the inclusion $(B\mathbb{G}_m)^2 \hookrightarrow (\mathbb{A}^1/\mathbb{G}_m)^2$.

The complement $|\mathrm{P}\overline{\mathcal{M}}_g^{\mathcal{B}}|^c$ is the union of the images under these gluing morphisms of two kinds of subsets of the product: 1) the locus of points where one marked point lies on a stable component and $\mathrm{ord}(s) > 1$ there, and 2) the locus of points where both marked points lie on a stable component, and $\mathrm{ord}_{p_1}(s_1) + \mathrm{ord}_{p_2}(s_2) \neq \deg(\mathcal{L})$. The condition that the marked point lies on a stable component is open, and the function $\mathrm{ord}_p(s)$ is constructible there, so the first kind of subset is locally constructible. Likewise, the second kind of subset is locally constructible because $\deg(\mathcal{L})$ is a constructible function on $\mathcal{X}_{0,2}$. The local constructibility of $|\mathrm{P}\overline{\mathcal{M}}_g^{\mathcal{B}}|^c$, and therefore the local constructibility of $|\mathrm{P}\overline{\mathcal{M}}_g^{\mathcal{B}}|$, then follows from the fact that the gluing morphisms are of finite type. □

3.3 Deformation theory of relatively polarized curves

Observation 3.3.1. Consider the forgetful morphism

$$F : \mathrm{P}\overline{\mathcal{M}}_g \to \mathfrak{M}_g^{tw,ss}, \quad (\mathcal{C}, \mathcal{L}, s : \mathcal{L} \to \mathcal{O}_{\mathcal{C}}) \mapsto \mathcal{C}.$$

By the proofs of Proposition 2.1.3 and Proposition 3.2.3, $\mathrm{P}\overline{\mathcal{M}}_g$ is an open substack of the mapping stack $\mathrm{Map}_{\mathfrak{M}_g^{tw,ss}}(\mathcal{C}_{univ}, \mathbb{A}^1/\mathbb{G}_m)$, where $\mathcal{C}_{univ} \to \mathfrak{M}_g^{tw,ss}$ denotes the universal family of twisted semistable curves and $\mathbb{G}_m$ acts linearly on $\mathbb{A}^1$ with weight -1. This mapping stack has a natural derived enhancement, given by the evident extension of the moduli functor to derived schemes as in [HLP23, §5.1]. Therefore, $\mathrm{P}\overline{\mathcal{M}}_g$ also admits a natural derived enhancement, which we denote by $\mathrm{P}\overline{\mathcal{M}}_g$ by a small abuse of notation. The description of the (co)tangent complex $\mathbb{T}_{\mathrm{P}\overline{\mathcal{M}}_g/\mathfrak{M}_g^{tw,ss}}$ in [HLP23, Prop. 5.1.10] implies that for all field-valued points $p = (\mathcal{C}, \mathcal{L}, s : \mathcal{L} \to \mathcal{O}_{\mathcal{C}})$ we have

$$p^*\mathbb{T}_{\mathrm{P}\overline{\mathcal{M}}_g/\mathfrak{M}_g^{tw,ss}} \cong R\Gamma\left(\mathcal{C}, [\mathcal{O}_{\mathcal{C}} \xrightarrow{s^\vee} \mathcal{L}^\vee]\right),$$

where we are taking the hypercohomology of a two term complex of sheaves on $\mathcal{C}$ with $\mathcal{O}_{\mathcal{C}}$ in cohomological degree -1. It follows in particular that the derived stack $\mathrm{P}\overline{\mathcal{M}}_g$ is quasi-smooth over the smooth stack $\mathfrak{M}_g^{tw,ss}$. In view of Lemma 3.1.5, then we have that $\chi(p^*\mathbb{T}_{\mathrm{P}\overline{\mathcal{M}}_g/\mathfrak{M}_g^{tw,ss}}) = 0$ for all points as above.

Proposition 3.3.2. *The following hold:*

(1) The stabilization morphism $\mathrm{st} : \mathrm{P}\overline{\mathcal{M}}_g \to \overline{\mathcal{M}}_g$ *is syntomic. In particular,* st *is flat and* $\mathrm{P}\overline{\mathcal{M}}_g$ *is lci.*

(2) The stack $\mathrm{P}\overline{\mathcal{M}}_g$ *is geometrically integral.*

In particular, for all $\mathcal{B}$ *as in Hypothesis 1.0.5, the open substack* $\mathrm{P}\overline{\mathcal{M}}_g^{\mathcal{B}} \subset \mathrm{P}\overline{\mathcal{M}}_g$ *also satisfies (1) and (2) above.*

Proof. After extending the base field, we may assume that k is algebraically closed.

(1). By Observation 3.3.1, $\mathrm{P}\overline{\mathcal{M}}_g$ admits a quasi-smooth derived enhancement relative to the smooth stack $\mathfrak{M}_g^{tw,ss}$, and hence the derived enhancement is (absolutely) quasismooth. Furthermore, the virtual rank of $\mathbb{T}_{\mathrm{P}\overline{\mathcal{M}}_g/\mathfrak{M}_g^{tw,ss}}$ is 0 at every point. By the distinguished triangle

$$\mathbb{T}_{\mathrm{P}\overline{\mathcal{M}}_g/\mathfrak{M}_g^{tw,ss}} \to \mathbb{T}_{\mathrm{P}\overline{\mathcal{M}}_g} \to F^*\mathbb{T}_{\mathfrak{M}_g^{tw,ss}},$$

the virtual rank of $\mathbb{T}_{\mathrm{P}\overline{\mathcal{M}}_g}$ at any point agrees with the virtual rank of $\mathbb{T}_{\mathfrak{M}_g^{tw,ss}}$, which is $3g-3$. Note that this agrees with the dimension of the smooth integral stack $\overline{\mathcal{M}}_g$.

To show that the morphism $\mathrm{st} : \mathrm{P}\overline{\mathcal{M}}_g \to \overline{\mathcal{M}}_g$ is syntomic, it suffices to show that the fibers of $\mathrm{st} : \mathrm{P}\overline{\mathcal{M}}_g \to \overline{\mathcal{M}}_g$ have dimension at most 0. We have a factorization $\mathrm{st} : \mathrm{P}\overline{\mathcal{M}}_g \xrightarrow{f} \mathfrak{M}_g^{tw,ss} \xrightarrow{\mathrm{st}} \overline{\mathcal{M}}_g$. The morphism $\mathrm{st} : \mathfrak{M}_g^{tw,ss} \to \overline{\mathcal{M}}_g$ is syntomic of relative dimension 0, and for all $p \in \overline{\mathcal{M}}_g(k)$ the locus $\mathfrak{M}_g^{tw,ss,=\ell} \cap \mathrm{st}^{-1}(p) \hookrightarrow \mathrm{st}^{-1}(p)$ where we have ℓ stacky rational bridges is locally closed of codimension ℓ in $\mathrm{st}^{-1}(p)$. It therefore suffices to show that the preimage $f^{-1}(x)$ of any point $x \in \mathcal{M}_g^{tw,ss,=\ell}(k)$ has dimension at most ℓ.

Fix such a point $x = \mathcal{C}$, and let $\mathcal{R}_1, \dots \mathcal{R}_\ell$ denote its rational bridges. Choose an orientation (p_-^i, p_+^i) for every bridge $\mathcal{R}_i$. Let $\widetilde{\mathcal{C}}^{\mathrm{st}} \subset \mathcal{C}$ denote the closure of the complement of the rational bridges as in Lemma 3.1.4. Each k-point $(\mathcal{C}, \mathcal{L}, s)$ in $f^{-1}(x)$ determines a tuple of types $\sigma := (\mathrm{type}(\mathcal{R}_i))_{i=1}^{\ell}$, where each $\mathrm{type}(\mathcal{R}_i)$ is a subset of $[-1,1] \cap \mathbb{Q}$. By Lemma 3.1.4, the denominators of the elements of $\mathrm{type}(\mathcal{R}_i)$ in reduced fraction form are uniquely determined by the isotropy groups ocurring in the oriented bridge $\mathcal{R}_i$, Hence there are finitely many possibilities for the collections of types σ, and we have a finite disjoint union decomposition $f^{-1}(x) = \sqcup_\sigma f^{-1}(x)_\sigma$ into constructible subsets. We may replace $f^{-1}(x)$ with a fixed $f^{-1}(x)_\sigma$; we need to show that its dimension is bounded by ℓ. By Lemma 3.1.4(1), for any geometric point the restriction $\mathcal{L}|_{\widetilde{\mathcal{C}}^{st}}$ is isomorphic to the ideal sheaf $\mathcal{O}(-\mathcal{D})$ of an effective Cartier divisor $\mathcal{D} \subset \widetilde{\mathcal{C}}^{st}$ which is uniquely determined by σ. Furthermore, the isomorphism classes of the restrictions $\mathcal{L}|_{\mathcal{R}_i}$ are also uniquely determined σ by Lemma 3.1.4(3). The possible choices of gluing data for obtaining $\mathcal{L}$ in terms of $\mathcal{O}(-\mathcal{D})$ and the line bundles $\mathcal{L}|_{\mathcal{R}^{(i)}}$ are parameterized by $\mathbb{G}_m^{2\ell}/\mathbb{G}_m^{\ell} \times \mathbb{G}_m^r \cong \mathbb{G}_m^{\ell-r}$, where r is the number of connected components of $\widetilde{\mathcal{C}}^{\mathrm{st}}$. Furthermore, by Lemma 3.1.4(2), the cosection is given as the composition

$$\mathcal{L} \twoheadrightarrow \mathcal{L}|_{\widetilde{\mathcal{C}}^{\mathrm{st}}} \xrightarrow{\psi} \mathcal{O}_{\widetilde{\mathcal{C}}^{st}}(-\mathcal{D}) \hookrightarrow \mathcal{O}_{\mathcal{C}}$$

for some choice of isomorphism $\psi : \mathcal{L}|_{\widetilde{\mathcal{C}}^{\mathrm{st}}} \xrightarrow{\sim} \mathcal{O}_{\widetilde{\mathcal{C}}^{st}}(-\mathcal{D})$ of line bundles. Hence for any fixed pair $(\mathcal{C}, \mathcal{L})$ the space of possible cosections is parameterized by $\mathbb{G}_m^r$. We conclude that the dimension of each of the finitely many $f^{-1}(x)_\sigma$ is at most $\ell - r + r = \ell$, as desired.

(2). This follows from flatness of $\mathrm{st} : \mathrm{P}\overline{\mathcal{M}}_g \to \overline{\mathcal{M}}_g$, the fact that st is an isomorphism over the substack of smooth curves $\mathcal{M}_g \subset \overline{\mathcal{M}}_g$, and the integrality of $\overline{\mathcal{M}}_g$. □

3.4 The proof of Theorem 1.0.7

The proof will appear at the end of this subsection.

The next lemma applies to a $\mathbb{G}_m$-equivariant family $(\mathcal{C}, \mathcal{L}, s)$ in $\mathcal{A}nc_g$ over Y, where Y is one of the two schemes associated in Notation 2.2.3 to a DVR R. Since $\overline{\mathcal{M}}_g$ is separated with finite inertia, the composition $\mathcal{Y} = Y/\mathbb{G}_m \to \mathcal{A}nc_g \to \overline{\mathcal{M}}_g$

factors through $\mathcal{Y} \to \mathrm{Spec}(R)$ by [AHLH23, Lem. 3.6, Lem. 3.45]. Concretely, this means the stabilization of $\mathcal{C}$ is pulled back from a family $C^{\mathrm{st}} \to \mathrm{Spec}(R)$ in the sense that there is a morphism $\mathcal{C} \to C^{\mathrm{st}}$ such that $\mathcal{C} \to Y \times_{\mathrm{Spec}(R)} C^{\mathrm{st}}$ is the stabilization.

As in Notation 2.4.4, we use the notation (u, v) for the pair of parameters cutting out $\mathfrak{o} \in Y$. We write η_u and η_v for the generic points of Y_u and Y_v.

Lemma 3.4.1. *If the restriction to Y° is a family in $\mathrm{P}\overline{\mathcal{M}}_g$, then so is the whole family over $Y/\mathbb{G}_m$. Furthermore, let $\mathcal{R}_u, \mathcal{R}_v$ and $\mathcal{R}_\mathfrak{o}$ denote the rational bridges lying over a node $x \in C_0^{\mathrm{st}}$ in each of the three fibers over η_u, η_v and $\mathfrak{o}$. Then a rational model for C^{st} at x canonically orients the three bridges, and*

$$\mathrm{type}(\mathcal{R}_\mathfrak{o}) = \mathrm{type}(\mathcal{R}_u) \cup \mathrm{type}(\mathcal{R}_v).$$

Proof. First, we observe that the cosection s on the fiber over $\mathfrak{o}$ can only vanish on rational bridges. This is because the vanishing locus of s on the family $\mathcal{C} \to Y$ has codimension at most one at every point, so if it meets a non-bridge point in the $\mathfrak{o}$ fiber it must also meet a non-bridge point in some other fiber, which would contradict Lemma 3.1.1.

Choose a rational model around $\mathrm{Spec}(R) \to \mathbb{A}^1$ and $\varphi : \widehat{C^{\mathrm{st}}} \cong \widehat{P_R}$ around a node $x \in C_0^{\mathrm{st}}$. The condition that s only vanishes on rational bridges implies the condition of the Lemma 3.1.7, that $V(s) \setminus \pi^{-1}(Z)$ is closed. Therefore Lemma 3.1.7 provides a family of genus 0 twisted curves $\mathcal{P} \to Y$, which by construction is $\mathbb{G}_m$-equivariant because $\mathcal{C}$ is. For any $y \in Y$, if $\mathcal{C}_y \cap \pi^{-1}(Z)$ is non-empty, then it is a rational bridge $\mathcal{R}_y \subset \mathcal{C}_y$, and the orientation of P_0^{un} both orients $\mathcal{R}_y$ and identifies a distinguished component $\mathcal{P}_y^- \subset \mathcal{P}_y$. The condition that $\mathrm{ord}_+(\mathcal{R}_y) + \mathrm{ord}_-(\mathcal{R}_y) = \deg(\mathcal{L}|_{\mathcal{R}_y})$ is equivalent to the condition that $\deg(\mathcal{L}_\mathcal{P}|_{\mathcal{P}_y}) = 0$, so if it holds at one fiber of Y it holds at every fiber.

The condition that $\mathrm{ord}_\pm(\mathcal{R}_y) \leq 1$ is equivalent to the condition that $\deg(\mathcal{L}_\mathcal{P}|_{\mathcal{P}_y^\pm}) \geq -1$. Any connected proper subcurve $\mathcal{S} \subset \mathcal{P}_y$ that contains $\mathcal{P}_y^-$ can be characterized as the smallest connected subchain that contains both $\mathcal{P}_\mathfrak{o}^-$ and x for some uniquely determined node $x \in \mathcal{P}_\mathfrak{o}$. x must deform to a node in one of the other fibers, so without loss of generality let us assume $x' \in \mathcal{P}_u$ is a node that specializes to x. Then the original subcurve $\mathcal{S}$ is the flat limit of the smallest connected subcurve $\mathcal{S}' \subset \mathcal{P}_u$ that contains $\mathcal{P}_u^-$ and x'. Therefore, the set of degrees of subchains of this form in $\mathcal{P}_\mathfrak{o}^-$ is the union of the set of degrees of subchains of this form in $\mathcal{P}_u^-$ and $\mathcal{P}_v^-$. This simultaneously implies the upper bound on $\mathrm{ord}_\pm(\mathcal{R}_y)$ and the formula for $\mathrm{type}(\mathcal{R}_\mathfrak{o})$. □

Proof of Theorem 1.0.7. By Proposition 3.2.3, $\mathrm{P}\overline{\mathcal{M}}_g^{\mathcal{B}} \subset \mathcal{A}nc_g$ is open for all $\mathcal{B}$ as in Hypothesis 1.0.5. Since $\mathcal{A}nc_g$ is Θ- and S-complete by Theorem 2.2.7, Lemma 3.4.1 shows that $\mathrm{P}\overline{\mathcal{M}}_g^{\mathcal{B}}$ is Θ- and S-complete as well.

If $\mathcal{B}$ is finite, Lemma 3.1.4 implies that $\mathrm{P}\overline{\mathcal{M}}_g^{\mathcal{B}}$ is bounded, and hence it admits a separated good moduli space by [AHLH23, Thm. A], which we call X. Because $\overline{\mathcal{M}}_g \subset \mathrm{P}\overline{\mathcal{M}}_g^{\mathcal{B}}$, the induced composition on moduli spaces $\overline{M}_g \to X \to \overline{M}_g$ is the identity, hence $\overline{M}_g \to X$ is a closed immersion. By the integrality of $\mathrm{P}\overline{\mathcal{M}}_g^{\mathcal{B}}$ proven in Proposition 3.3.2, the open immersion $\overline{\mathcal{M}}_g \hookrightarrow \mathrm{P}\overline{\mathcal{M}}_g^{\mathcal{B}}$ is dense, and hence $\overline{M}_g \to X$ is also dense. This forces the closed immersion $\overline{M}_g \to X$ to be an isomorphism, and thus the same holds for $X \to \overline{M}_g$. □

A Formal gluing of algebraic stacks

In the proofs of Lemma 2.3.4 and Lemma 3.1.7 we use the following gluing result for algebraic stacks. A proof will appear in a future version of this paper.

Given an algebraic stack $\mathcal{X}$ and a closed subset $Z \subset |\mathcal{X}|$, a *modification of* $\mathcal{X}$ *over* Z is a morphism of algebraic stacks $f\colon \mathcal{Y} \to \mathcal{X}$ such that the restriction $f^{-1}(\mathcal{X} \setminus Z) \to \mathcal{X} \setminus Z$ is an isomorphism.

Proposition A.0.1. *Suppose $\mathcal{X}$ be an algebraic stack, $Z \subset |\mathcal{X}|$ is a closed subset, and $g\colon \mathcal{X}' \to \mathcal{X}$ is morphism of algebraic stacks that induces an isomorphism of formal completions along Z. Then base change along g is an equivalence between the 2-category of modifications of $\mathcal{X}$ over Z, and the 2-category of modifications of $\mathcal{X}'$ over $g^{-1}(Z)$. Furthermore, a modification $\mathcal{Y} \to \mathcal{X}$ is separated if and only if $\mathcal{Y} \times_{\mathcal{X}} \mathcal{X}' \to \mathcal{X}'$ is.*

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